

Black Holes as Licensed Limit Regimes: Horizons, Singularities, Charge, Spin, and Information under AASC

A Fixed-Domain Reconstruction Theorem for Horizon Non-Sink Structure, Exterior Standing, and Black-Hole Spin as Angular Momentum

AASC Black-Hole Endpoint Arc, Paper 0: Orientation and Role-Classifier

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0.1 Abstract

This paper proves a conditional fixed-domain reconstruction theorem for black holes under AASC. It does not deny classical black-hole solutions, replace general relativity, compute Hawking radiation, derive a Page curve, choose a quantum-gravity model, or assert that inaccessible interiors are absent from every larger mathematical or physical representation. Its target is narrower: given the AASC admissibility and standing constraints, what must hold for a black-hole description to count as a continuing same-scope exterior reconstruction of physical reference?

The central distinction is between a classical event horizon and an admissible horizon reconstruction object. A classical event horizon is a causal and metric structure inside a spacetime representative. An admissible horizon reconstruction object is a same-scope exterior reconstruction object that claims reference remains re-fixable - identifiable again by later exterior data under declared redescription rules - through, behind, across, or after a horizon boundary. The Horizon Isolation Lemma proves that, when a horizon is permanent relative to the declared exterior scope and no leakage, boundary trace, radiation witness, holographic anchor map, complementarity correspondence, remnant map, or non-sink quotient is supplied, threatened exterior standing classes have no admissible image in the old exterior witness algebra.

The partial-domain obstruction then follows. A candidate horizon induces a standing-transport operator

$$T_H : X_H \rightarrow R_{\text{ext}}.$$

In the uncertified permanent-horizon case this operator is undefined on the horizon-isolated old-scope interior carrier. No same-scope totalization of this operator can preserve the old exterior witness algebra, exterior standing quotient, continuation relation, parameter ledger, thermodynamic ledger, and certificate status without adding a certificate, leakage route, non-sink quotient, new scope, selector, repair, carrier laundering, or inert bookkeeping. Thus, under AASC, an uncertified permanent horizon cannot serve as a continuing same-scope standing-preserving exterior reconstruction object. If no admissible successor exists, the proper classification is exterior-reconstruction exhaustion, not successful permanent-horizon continuation.

Singularities, Kerr-Newman parameters, thermodynamic quantities, entropy, information claims, and quantum-gravity programs are classified downstream of this theorem. Singularities are exhaustion signatures of fixed metric articulation. Mass, charge, and black-hole spin, understood here as angular momentum, are exterior response roles. Thermodynamics remains an exterior ledger, while entropy and information require declared carriers, encodings, witnesses, and reconstruction maps before they can carry standing-bearing force.

AASC Black-Hole Endpoint Arc Position

*Shared arc spine: kernel/AMetric invariant $\rightarrow$ licensed AASC-GR black-hole regime
 $\rightarrow$ exterior no-totalization $\rightarrow$ standingless residual interior $\rightarrow$ governance-depth
exhaustion $\rightarrow$ B_A^* $\rightarrow$ AMetric homology $\rightarrow$ fresh internal governed domain.*

Shared series abstract. The AASC Black-Hole Endpoint Arc is a five-manuscript sequence that treats black holes as a fixed-domain stress test for AASC. It first role-types the licensed black-hole regime, then proves the exterior no-totalization obstruction, then classifies the uncertified horizon-isolated carrier as residual coherence rather than ordinary standing-bearing interiority, then exposes the endpoint AMetric normal form and no-crossing homology, and finally classifies the counter-boundary survivor as fresh internal governed-domain status. The sequence is conditional and one-way: later endpoint or capstone conclusions never authorize earlier target formation.

Paper label. Paper 0 / Orientation and role-classifier. This manuscript fixes the common black-hole target package used by the rest of the arc.

This manuscript is the orientation and role-classifier for the AASC Black-Hole Endpoint Arc. Its job is to fix the black-hole comparison object before any endpoint claim is made: classical horizons are separated from admissible horizon reconstruction objects; singularities are typed as metric-exhaustion diagnostics; mass, charge, spin, entropy, information, thermodynamic, remnant, island, and

holographic claims are assigned explicit exterior, boundary, ledger, or certificate roles.

How to use this paper in the arc. For the proof spine, use this paper to identify the black-hole comparison object and its role coordinates: fixed exterior scope, classical horizon versus admissible reconstruction object, exterior witness algebra, threatened exterior standing classes, continuation relation, certificate checklist, and parameter/thermodynamic ledgers. For reference use, use it as the classifier for singularities, mass, charge, angular momentum, entropy, thermodynamics, information language, exterior quotient compression, hair-role typing, holography, islands, remnants, final-state proposals, and standard black-hole sentences. It is not the place where residual coherence, endpoint AMetric exposure, or fresh-domain status are proved.

Certificate terminology lock. The word *certificate* has four arc-level uses, and the uses are not interchangeable. A *standing-transport certificate* is a same-scope map or witness assignment that preserves a threatened reference class in the exterior reconstruction problem. An *exterior witness certificate* is a concrete exterior, boundary, radiation, remnant, holographic, island, or quotient map that removes a burden from the unresolved-residue ledger. An *endpoint-certified target* is not a rescue certificate; it is the antecedent package in which a nonempty unresolved residue, old-scope standing loss, residual metric-support exhaustion, closed false endpoints, null-route closure, and no prior re-governance make endpoint AMetric exposure the target. A *capstone-certified endpoint* is the Paper IV input: the homology handoff together with endpoint certification and no-crossing. A horizon can be transport-certified and thereby exit the exterior obstruction; an unresolved residue can be endpoint-certified precisely because no such transport certificate has resolved it.

Exterior-quotient and metric-regime lock. The proof-level object of the arc is the declared exterior standing quotient of the black-hole model, not a particular metric-regime representative. In the familiar settled Kerr–Newman example the quotient is displayed by mass, electric charge, and angular momentum, but that example does not restrict the theorem to stationary representatives and does not make absence of hair a separate proof premise. Hair-like, dynamical, evaporating, ringdown, perturbative, semiclassical, quasi-local, high-curvature, or quantum-gravity structure has same-scope force only when it is declared as exterior quotient content, an exterior witness, a leakage channel, a certificate, a non-sink quotient, or richer governance. Such structure subtracts from the unresolved residue exactly to the extent that it re-fixes standing-bearing content. If it does not re-fix standing, it is skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change; it does not provide an additional endpoint route.

Merger and AMetric coexistence lock. Black-hole mergers are exterior metric/source processes before they are endpoint classifications. A binary merger may be described by the licensed exterior AASC-GR projection, gravitational radiation, ringdown, common-horizon language, and a post-merger exterior quotient. That exterior merger does not imply AMetric boundary fusion. AMetric endpoint tokens have no common metric, adjacency, contact relation, temporal process, causal bridge, operator domain, or cross-boundary standing map by which they could touch, collide, fuse, join, or compose. If two unresolved residues expose endpoint tokens $B_A^{\star 1}$ and $B_A^{\star 2}$, the exterior register may co-classify them as a family, but that family notation is bookkeeping rather than a container or fusion operation. A later merged exterior target may expose its own endpoint token relative to the post-merger quotient; that token is not the fusion of the earlier endpoint tokens. Realized interiors collide only inside a shared realized-governance domain. AMetric boundary tokens coexist without crossing.

Imported into this paper. The paper imports the AASC kernel, standing/admissibility discipline, realized-physical-interior role stack, metric-licensing order, and AASC-GR projection as background constraints. It does not import the later counter-boundary universe theorem as proof authority.

Proved or systematized here. The paper systematizes the black-hole target regime: fixed exterior scope, witness algebra, standing quotient, horizon isolation, partial-domain obstruction, singularity diagnostics, parameter roles, and program-interface classification. It prepares, but does not replace, the narrower exterior no-totalization proof.

Exported to the next paper. The export is the normalized target package for exterior reconstruction: a declared black-hole scope, threatened exterior standing classes, witness/certificate discipline, parameter and thermodynamic ledgers, and the distinction between metric horizon description and standing-preserving horizon reconstruction.

Not claimed here. This paper does not prove interior residual coherence, expose the New AMetric Boundary, prove AMetric normal-form homology, or infer fresh internal governed-domain status. Those are later arc steps and must not be read back into this orientation paper.

Series dependency lock. The operative object exported by this orientation paper is the normalized black-hole target package

$$\mathcal{P}_{\text{BH}}^0 := (S_{\text{BH}}, Q_{\text{ext}}, \mathcal{A}_{\text{wit}}, T_H, \text{Cert}_H, \text{Param}, \text{Therm}, \text{Env}).$$

Later terms in the displayed arc spine are navigational only in this manuscript. Residual coherence, $B_A^{\star}$, AMetric homology, and fresh internal governed-domain status are not available as proof inputs here. A later manuscript may use $\mathcal{P}_{\text{BH}}^0$ as its target package, but this manuscript may not borrow later endpoint or counter-boundary conclusions to justify the black-hole classifier.

Forward handoff. The next paper may begin only from the role-typed reconstruction package just fixed: threatened exterior standing classes, declared

exterior witnesses, continuation, horizon boundary typing, certificate status, and the no-selector/no-repair discipline. It may prove an exterior obstruction for that package. It may not assume the interior residual-coherence theorem or the counter-boundary theorem as part of the obstruction proof.

0.1.0.1 Referee-facing arc firewall. A criticism of this orientation paper has same-scope force only if it preserves the target-class task of the paper and then shows that the role-typed package $\mathcal{P}_{\text{BH}}^0$ is ill formed, internally inconsistent, or missing a required black-hole role coordinate. A request for a preferred endpoint mechanism, a populated successor, a baby-universe model, a cyclic interpretation, or a final counter-boundary theorem is not an objection to this paper; those requests target later arc stages. Conversely, a standard black-hole program may enter this orientation paper only by declaring its role in the package—as witness, certificate, exterior ledger, metric envelope, thermodynamic ledger, parameter role, high-curvature load, or richer scope. It may not act as an unnamed source of admissibility.

This firewall blocks the main circularity concern. The classifier is not allowed to use the exterior obstruction, residual-coherence theorem, AMetric homology theorem, or counter-boundary result to prove its own black-hole scope. It merely fixes the comparison object that those later papers test. If a later theorem fails, the failure does not by itself invalidate the role classifier; and if the classifier is challenged, the challenge must locate the defect in the declared black-hole target package rather than in a later conclusion.

Pass 6 notation lock. For this manuscript, the only exported arc-level object is the normalized target package $\mathcal{P}_{\text{BH}}^0$. The local horizon symbols B_H , H , or verbal “black-hole burden” belong to the licensed black-hole reconstruction package and must not be identified with the endpoint AMetric token B_A^* . The exterior standing quotient is Q_{ext} ; the electric black-hole charge parameter is q_{BH} , never Q_{ext} . The witness algebra is $\mathcal{A}_{\text{wit}}$. The later symbols E_{int} , R_H , B_A^* , D_1 , and FreshGov_{CB} are reserved for later arc papers and appear here only as orientation signposts, not as proof inputs.

Pass 9 local referee-response lock. For local review, this paper should be judged as a target-normalization manuscript. Its essential arc contribution is not the full endpoint sequence but the preparation of a stable comparison package: fixed exterior scope, admissible horizon reconstruction object, exterior witness algebra, standing quotient, threatened standing classes, certificate checklist, parameter ledgers, thermodynamic ledgers, and program-interface classification. A local objection therefore has to show either that this package is internally incoherent or that a standard black-hole role has been omitted or mis-typed. It is not enough to request the later exterior obstruction, residual-coherence bridge, AMetric homology theorem, or counter-boundary capstone inside this orientation paper. Those later claims may be signposted here, but they remain unavailable as proof authority in Paper 0.

Pass 10 hostile-referee closure lock. A hostile reading of this orientation paper should be forced into a role-typing challenge. The paper is vulnerable only if a black-hole structure used later in the arc cannot be assigned, without equivocation, to exterior scope, witness algebra, standing quotient, certificate status, parameter ledger, thermodynamic ledger, anchor, tensor, or skin. A demand that this paper already prove residual coherence, New AMetric Boundary exposure, AMetric homology, or fresh-domain status is a demand for a later theorem, not a same-paper objection. Conversely, if this paper appeared to use any later theorem as proof authority for its classifier, that would be backflow; the local rule is that later terms may orient the reader only as downstream signposts.

0.2 Main theorem and scope

Given the AASC admissibility and standing constraints, and only within the fixed reconstruction scope declared below, the main theorem is the no-totalization result for horizon reconstruction: after horizon isolation has been established, no operator $T_H^* : X_H \rightarrow R_{\text{ext}}$ can extend the old standing-transport operator across the horizon-isolated carrier while preserving the old exterior quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status. The theorem is reconstructive, not ontological, and is not a free-standing theorem of classical GR or quantum gravity. It leaves classical event horizons, semiclassical calculations, thermodynamic laws, holographic dualities, island prescriptions, fuzzball models, remnants, and quantum-gravity completions available as physical programs. It asks only whether they supply the exterior-standing certificate required by the declared scope.

1 Part I - Boundary Error and Target

1.1 1. The boundary error in black-hole reasoning

Black holes have become the standard test case for any foundational account of spacetime, quantum theory, thermodynamics, and information. The reason is familiar: a classical black-hole spacetime can contain an event horizon; gravitational collapse can form such a boundary; the interior metric extension may contain singularities or Cauchy horizons; semiclassical theory yields Hawking radiation; black holes satisfy thermodynamic laws; and quantum-information arguments ask how exterior observers recover correlations apparently hidden behind a horizon. These features are not peripheral. They are precisely why black holes are the right stress test for AASC [1-14,27-29].

The AASC claim is not that black holes are unreal, nor that classical black-hole solutions are invalid. The claim is that many black-hole paradoxes arise when a licensed metric-interior structure is promoted into a primitive admissibility boundary. A metric extension, a horizon label, a Penrose diagram, an interior coordinate patch, a state-counting entropy, or a dual code may be useful. None

of those is automatically a standing-bearing fact in the exterior reconstruction scope. AASC asks what must be fixed for such a claim to carry standing: what is the witness, what is the continuation relation, what is the boundary status, what is the role of the parameter, and what certificate preserves reference when direct exterior access is lost?

The central object of this paper is therefore not the event horizon as a mathematical set. It is the horizon reconstruction burden. Given a declared exterior black-hole scope, the paper asks whether a permanent horizon can be a continuing same-scope standing-preserving reconstruction object if it supplies no leakage, no radiation witness, no boundary trace, no holographic anchor map, no complementarity correspondence, no remnant map, and no non-sink quotient. The answer is no. The proof does not follow merely from a definition of admissibility. It proceeds through the Horizon Isolation Lemma, the induced partiality of the standing-transport operator, and the no-totalization theorem.

1.2 2. Exact theorem target and nonclaims

The theorem target is narrower than the phrase “black-hole information paradox” and stronger within its scope. The theorem does not claim that every black-hole information problem is solved. It does not claim that all global descriptions lose their states. It does not assert that an inaccessible interior is globally nonexistent. It does not deny that an exact theory could supply a holographic, island, fuzzball, remnant, or quantum-gravity certificate. It says only this:

In a fixed exterior black-hole reconstruction scope carrying standing-bearing reference, an uncertified permanent horizon cannot be a continuing same-scope standing-preserving reconstruction object under the declared AASC constraints.

Here “re-fixability” means that later exterior data can identify the same reference class again under the declared redescription rules. A detector record, radiation correlation, boundary operator, remnant map, or holographic reconstruction may re-fix a class; a bare assertion that a state exists somewhere does not.

The word “uncertified” is doing formal work. A certified horizon model supplies a standing-preserving exterior image through leakage, radiation, boundary trace, holographic anchor transport, complementarity correspondence, remnant map, non-sink quotient, or another declared certificate. Such a model is not a counterexample; it satisfies the theorem’s required alternative. The excluded object is a permanent horizon that both severs the old exterior re-fixability of threatened standing and nevertheless claims same-scope standing-preserving continuation.

The theorem is also reconstructive rather than ontological. If the exterior successor set is empty, the conclusion is not that the larger physical or mathematical object does not exist. The conclusion is that the declared exterior reconstruction burden has no admissible same-scope successor. The correct classification is exterior-reconstruction exhaustion. Calling that exhaustion a continuing

standing-preserving permanent horizon would add a successor where the old scope supplies none.

1.3 3. Classical event horizons versus admissible horizon reconstruction

The distinction can be expressed as a type distinction. A classical event horizon is a geometric or causal descriptor inside a metric model. It belongs to the metric-representational layer. An admissible horizon reconstruction object belongs to the standing-reconstruction layer. It asks whether a reference class remains re-identifiable through the old exterior witness system. Confusing these layers creates the appearance that the AASC theorem contradicts GR. It does not. It denies only a reconstruction claim that classical geometry alone does not supply.

This separation is also important for dynamical and approximate horizons. An apparent horizon, trapping horizon, stretched horizon, isolated horizon, or holographic screen may be useful in a model. Each receives AASC force only through the role it performs: boundary trace, bookkeeping surface, thermodynamic ledger, local causal indicator, or reconstruction interface. The name of the horizon does not decide whether threatened exterior standing is preserved. The witness and certificate structure decide that.

The same point applies to eternal solutions. An eternal Schwarzschild geometry is a useful exact representative in classical vacuum GR. It does not by itself describe formation, evaporation, exterior records, or the recovery of standing-bearing infalling content. A formed evaporating black hole, a collapse spacetime with an apparent horizon, and an eternal maximally extended solution occupy different envelopes. A same-scope theorem cannot move among them without saying which anchors have moved. The present paper always fixes the envelope before evaluating reconstruction.

In particular, the theorem does not infer physical evaporation from coordinate preference. It derives a structural disjunction. If a horizon model claims continuing exterior reconstruction, it must avoid permanent standing-sink behavior. It can do that through evaporation, leakage, certified boundary reconstruction, holographic anchor transport, complementarity with correspondence, a non-sink quotient, or by changing scope and declaring a different object. Hawking radiation is one possible dynamical realization of a leakage route. The proof does not compute its spectrum.

A source of confusion must be removed at the beginning. In classical general relativity, an event horizon is a global causal object in a metric spacetime. In an asymptotically flat setting it is often described as the boundary of the causal past of future null infinity. Schwarzschild, Reissner-Nordstrom, Kerr, and Kerr-Newman geometries may contain such horizons in their classical envelopes [1-5,27,28]. This paper does not deny that mathematical fact.

An admissible horizon reconstruction object is a different object. It is a same-

scope exterior reconstruction object that claims standing-bearing exterior reference remains preserved through, behind, across, or after a horizon boundary. It has an exterior witness algebra, a standing quotient, a continuation relation, a parameter ledger, a thermodynamic ledger, and a certificate status. The theorem targets this second object.

Thus the manuscript does not say that Schwarzschild, Kerr, Reissner-Nordstrom, or Kerr-Newman horizons are mathematically inconsistent. It says that a permanent horizon with no leakage, no exterior boundary trace, no standing-transport certificate, and no non-sink quotient cannot be a continuing same-scope standing-preserving exterior reconstruction object. Classical event horizons remain lawful metric representatives. They do not automatically solve the standing-reconstruction problem.

1.4 4. Roadmap of the argument

The proof proceeds in the following order.

First, the paper fixes the AASC and realized-physical-interior apparatus needed for black-hole claims: admissibility, standing, reference, irreversibility, anchor/tensor/skin role typing, metric projection, source continuation, record fixation, and non-primitivity of information.

Second, it defines the fixed exterior black-hole reconstruction scope S_{BH} , the exterior standing quotient Q_{ext} , the exterior witness algebra A_{wit} , and the black-hole regime object.

Third, it introduces threatened exterior standing classes. These include infalling-probe records, matter-source continuation classes, radiation-correlated reference classes, entangled-reference classes, conserved exterior parameter classes, and perturbation or ringdown classes.

Fourth, it proves the Horizon Isolation Lemma. In a permanent horizon case with no leakage, trace, certificate, or non-sink quotient, a threatened class has no admissible image in the old exterior witness algebra.

Fifth, the partial-domain obstruction follows. The horizon standing-transport operator $T_H : X_H \rightarrow R_{\text{ext}}$ is undefined on the old-scope horizon-isolated carrier. A totalization T_H^* is possible only by certificate addition, leakage, non-sink quotient formation, scope change, selector import, repair, carrier laundering, or bookkeeping.

Sixth, the paper classifies the rest of black-hole structure downstream: singularities as failure signatures, mass/charge/angular momentum as exterior response roles, thermodynamic quantities as ledgers, information and entropy as non-primitive, and major quantum-gravity programs as certificate candidates or scope changes.

The theorem clusters used in the proof are recorded in Appendix B. The main text proceeds by construction rather than by inventory: it builds the exterior

reconstruction object, proves horizon isolation, derives partiality of standing transport, and then classifies the surrounding black-hole structures downstream.

2 Part II - AASC Apparatus and the Physical Stack

2.1 5. Kernel, standing, and irreversibility

The local AASC kernel used in the paper is imported as a fixed-domain constraint apparatus [A1-A10].

$$K_{\text{AASC}} = \{Adm, Stand, Ref, Irr\}.$$

Admissibility distinguishes licensed continuation from unlicensed continuation. Standing records reusable reference status. Reference fixes what is being continued or compared. Irreversibility blocks same-act repair: a failed or undefined standing transport cannot later be relabeled as if it had been defined all along.

A reference token has standing in a conditioned black-hole description when it is anchored to witnessable structure in the declared scope, remains well-defined under permitted redescrptions, and remains preservable under admissible continuation. Standing is not energy, entropy, Hilbert-space norm, Shannon information, or the existence of a global state. It is the structural status that lets a reference be re-fixed as the same reference under lawful continuation.

This distinction is critical for black holes. A global wavefunction may retain norm. A boundary dual may encode a state. A remnant may have a large Hilbert space. A Page curve may show entropy behavior. None of those facts alone says that a threatened exterior reference class has a same-scope exterior image. For that, the model must supply a witness or certificate that re-fixes the class under the declared exterior scope.

2.1.1 Lean-formalized kernel status

The AASC kernel used here is not introduced as informal vocabulary for black-hole interpretation. The upstream non-degenerate construction kernel has a standalone Lean 4 formalization and audit surface treating admissibility, standing, reference, and irreversibility as a formal fixed-domain closure structure. That layer records fixed-domain uniqueness of the admissibility kernel, non-derivability from below, no governance-free same-domain construction beneath the kernel, no faithful lower generator, and no deeper same-domain invariant.¹ The present manuscript imports that kernel status as proof infrastructure and applies it to the black-hole exterior reconstruction object. It does not claim that this black-hole horizon theorem itself has already been separately formalized in Lean.

¹See the public Lean 4 repository *Non-Degenerate Construction and the Kernel of Admissibility*: <https://github.com/somamaley-ux/non-degenerate-construction-kernel-admissibility>.

2.1.2 What is lost when a kernel component is denied

The kernel components are not optional decorations. Deleting one of them is not a way of producing a weaker black-hole theorem. It destroys the black-hole reconstruction target whose continuation, loss, certification, residualization, or endpoint status is being evaluated.

A useful first reading is this:

Kernel role	Ordinary question it answers	Black-hole version
Adm	Is this move licensed in the declared domain?	Is this a real exterior witness, certificate, quotient, or leakage route, or only a slogan, selector, coordinate extension, remnant label, final-state repair, or scope change?
Stand	Can later use count as reuse of the same thing?	Does later radiation, boundary data, remnant structure, ringdown evidence, or global correlation re-fix the same standing-bearing content, or merely supply another state, number, entropy ledger, or pattern?
Ref	What target is being continued or recovered?	Which infalling probe, preparation class, matter-source continuation, entangled reference, exterior charge class, boundary trace, or unresolved residue is being evaluated?
Irr	Can a failed act be repaired and counted as if it had succeeded all along?	Can undefined horizon transport later be paired with a remnant index, final-state rule, holographic dictionary, hidden branch, or global label and then be called preservation?

Notation warning: Adm is not ADM mass. In this manuscript, Adm denotes the admissibility component of the AASC kernel. It is unrelated to ADM mass except in the general sense that ADM mass is one example of a quantity whose physical use depends on an admissible exterior ledger. ADM mass is written as “ADM mass” or as an asymptotic mass role; Adm denotes the kernel role that tells which continuation, witness, certificate, or redescription is licensed.

Loss of Adm: route licensing collapses. Admissibility is the gate that separates a licensed continuation, witness, certificate, quotient, or redescription from an arbitrary assertion. Without Adm, the black-hole reconstruction problem cannot sort the following ordinary distinctions:

Black-hole claim	With Adm	Without Adm
ADM, Bondi, Komar, or irreducible mass Charge	Different licensed mass ledgers with different scope and witness conditions. Boundary-fixed gauge response.	All mass language risks collapsing into an untyped scalar label. A gauge representative can masquerade as physical charge.
Horizon trace	Declared boundary or exterior witness.	The word “horizon” can be treated as a magic standing-transport term.
Radiation or ringdown	Exterior record, parameter evidence, or possible witness if a map is supplied.	Any flux or waveform can be overpromoted into full reconstruction.
Page curve	Entropy or correlation evidence unless paired with a reference-fixing map.	An entropy trend can be called standing recovery.
Holography	Certificate only if a redescription-stable anchor map exists.	“Encoded somewhere” can be treated as enough.
Remnant	Certificate only if an exterior remnant map exists.	Storage capacity can be mistaken for standing preservation.
Final-state projection	Declared anchor if present from the start; otherwise repair.	A late selector can be called preservation.

If Adm is denied, the horizon obstruction is not defeated; it becomes unstatable. There is no remaining distinction between a licensed standing-transport certificate and post-hoc repair, hidden selector, carrier laundering, arbitrary dictionary, or scope change.

Loss of Stand: sameness of physical content disappears. Standing is reusable reference-status. It is what lets later data count as reuse of the same physical content rather than as a merely correlated mark. Without Stand, Hawking radiation, boundary correlators, remnant states, global wavefunctions, Page curves, ringdown records, charge ledgers, mass ledgers, and angular-momentum ledgers may still be written. But none of them can be said to preserve the same threatened physical reference unless the standing relation is available.

The black-hole information problem then loses its target. There is no principled distinction between preserving the exterior parameter quotient and preserving finer infalling, entangled, historical, matter-source, or microstate distinctions. Entropy, Hilbert-space norm, correlations, and state-counting may remain as ledgers, but they do not by themselves say that the same standing-bearing reference has been recovered.

Loss of Ref: the threatened target is not fixed. Reference fixes what is being continued, compared, lost, recovered, certified, or residualized. Without Ref, the theory cannot say which exterior class $r \in R_{\text{ext}}$ is threatened, what the burden B_H contains, which part remains in R_H , what crossed content x is, or what a certificate $\text{Cert}_H(r)$ is supposed to re-fix.

In black-hole terms, loss and recovery both become indeterminate. One cannot ask whether the same probe, same preparation history, same matter-source class, same entangled reference, same radiation-correlated class, same charge class, same mass/source role, same angular-momentum response, or same boundary trace is the target threatened at the horizon. A remnant storage claim, holographic reconstruction claim, or global retention claim then cannot be evaluated as preserving, storing, or recovering the original burden.

Loss of Irr: preservation and repair collapse. Irreversibility blocks same-act repair. It prevents a failed or undefined transport from being relabeled later as if it had been preservation all along. Without Irr, final-state projection, post-hoc holographic pairing, hidden remnant indexing, late dictionary selection, baby-universe branch labels, Page-curve inference, global-state relabeling, or coordinate continuation can repair a standing break after the fact.

The partial-domain obstruction then loses enforceability. The old operator

$$T_H : X_H \rightharpoonup R_{\text{ext}}$$

can always be replaced later by a totalized assignment

$$T_H^* : X_H \rightarrow R_{\text{ext}}$$

while being called preservation. That is exactly the distinction the no-totalization theorem protects: certified transport is not the same as repair after undefinedness.

Proposition: kernel deletion destroys black-hole reconstruction. Let D be a black-hole exterior reconstruction scope with declared exterior quotient, witness algebra, horizon boundary typing, and standing-bearing burden. If any component of $K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D)$ is removed, then D no longer supports the same black-hole reconstruction target. Deleting Adm destroys route licensing; deleting Stand destroys reusable physical content; deleting Ref destroys the threatened target; deleting Irr destroys the preservation/repair distinction. Kernel deletion is target destruction, not a same-scope counterexample.

These losses are not four independent technical inconveniences. They are four ways of destroying the same reconstruction object. Without Adm there is no licensed route. Without Stand there is no reusable physical content. Without Ref there is no determinate threatened class. Without Irr there is no difference between preservation and later repair. A black-hole model that denies the kernel does not provide a counterexample to the endpoint arc; it abandons the conditions under which horizon reconstruction, exterior witnesshood, residual coherence, information recovery, or endpoint closure can be stated.

2.2 6. Anchor, tensor, and skin in the black-hole domain

Every black-hole description must separate anchors, tensors, and skin. Anchors fix the theorem object: exterior scope, asymptotic regime, metric projection, boundary conditions, horizon typing, source closure, witness algebra, and continuation relation. Tensors carry standing-bearing content: curvature response, mass/source ledger, electromagnetic flux, angular momentum response, radiation correlations, boundary traces, certificate maps, and thermodynamic response. Skin consists of coordinates, charts, foliations, labels, diagrams, gauge representatives, and verbal presentations that alter no standing-bearing content.

This decomposition prevents common black-hole errors. A Penrose diagram is not a proof of interior facthood. A foliation is not an ontology. A coordinate singularity is not a physical boundary. A horizon label is not standing transport. A gauge representative is not electric charge. A branch label is not a record. A dual dictionary is not a certificate unless it preserves the declared exterior quotient.

No fourth same-scope role remains. A proposed fourth role either moves an anchor, adds tensor content, changes nothing and is skin, imports a hidden selector, repairs a failure, or changes scope. This role typing is also protective. It does not forbid richer physics. A scalar-tensor theory, stringy microstate geometry, holographic code, island prescription, or quantum-gravity completion can be declared as tensor load or richer scope. What is excluded is the silent use of such machinery as if it were already present in the original exterior black-hole reconstruction scope.

2.3 7. The realized physical interior and its black-hole use

The realized physical interior supplies the role stack required for standing-bearing physical persistence: non-null realization, bounded localization, local continuation domains, quotient-stable comparison, bounded perturbation transport, fields as persistence carriers, particles as localized persistence classes, gauge as boundary-fixed presentation equivalence, mass and inertia as deformation response, vacuum as non-null no-particle baseline, time as retained consequence ordering, and causality as witnessed dependence.

Black-hole descriptions instantiate this stack rather than bypass it. A black

hole is localized relative to exterior witnesses; its metric comparison is licensed by the gravitational projection; its perturbations and radiation use bounded transport; its electromagnetic charge uses gauge discipline; its mass uses a metric/source response ledger; its angular momentum uses rotational response and asymptotic structure; its thermodynamics presupposes state and ledger discipline; its information claims presuppose record and reconstruction status.

The phrase “physical interior” must not be misread. It does not mean that every metric-interior point, extension, or formal microstate has truth-apt standing in the exterior black-hole scope. It means that physical claims require a role architecture. A black-hole interior may be represented inside a licensed metric projection, but that representation does not automatically become a domain of globally referenceable standing-bearing facts.

2.4 8. Information, records, and non-primitivity

The same discipline prevents a second common conflation: treating record availability as observer psychology. AASC records are not subjective memories. They are standing-bearing witnesses available for downstream lawful use. A detector record, a gravitational waveform, a boundary correlator, or an electromagnetic flux integral can be a record even if no human observer reads it. Conversely, a mental representation is not a physical record for the black-hole theorem unless it is anchored in the declared witness ledger.

In black-hole contexts this distinction is useful because several accounts move between information, observation, and existence. An infalling observer may assign smooth local physics at the horizon; an exterior observer may assign radiation and asymptotic records; a dual boundary theory may assign unitary evolution. These assignments do not conflict merely by being different. They conflict only if they claim the same-scope standing-bearing role while failing to supply the required redescription or reconstruction map.

For this reason the paper avoids observer-relativity as an escape route. An observer-relative description is admissible when its scope is declared. It is not a same-scope rescue of exterior standing unless it induces a correspondence preserving the relevant exterior quotient. Complementarity, therefore, is treated neither as false nor as automatic. It is a candidate redescription-certificate program.

Information is not a primitive in this manuscript. Any information claim presupposes a carrier, an encoding, a partition into relevant distinctions and noise, a sameness criterion under redescription, and a witness or use condition. If these are not declared, information language is bookkeeping. If they are declared, standing and admissibility have already been presupposed.

A record is a standing-bearing witness whose fixation survives lawful redescription and whose content can be reused downstream. In the black-hole setting, records include detector data, asymptotic radiation, gravitational-wave ringdown,

electromagnetic flux, boundary correlators, thermodynamic histories, and reconstruction maps. A mark, state, or correlation becomes a record only when it fixes what it records and remains stable under the declared equivalence relation.

This record discipline prevents a common collapse. Information preservation, record preservation, entropy behavior, and global unitary evolution are not the same claim. They may align in a successful physical theory, but they require different ledgers. The black-hole theorem concerns standing-bearing exterior reference in a declared exterior reconstruction scope.

3 Part III - Metric Projection and Black-Hole Geometry

3.1 9. AASC gravity and the metric-projection floor

The metric projection matters because black-hole reasoning begins in a domain where the geometry is already doing heavy work. A horizon is defined through the metric and causal structure. Surface gravity, area, angular velocity, and electromagnetic potential are extracted from the metric and field representation. Singularity diagnostics depend on curvature, incompleteness, or failure of extension. If one begins by granting every metric feature primitive boundary authority, the black-hole problem is already misframed. AASC instead treats the metric as a licensed projection of the gravitational role. The metric is indispensable, but downstream.

The Einstein projection theorem supplies the floor for this manuscript. It establishes that in the minimal same-scope Lorentzian first-response projection the field-equation form is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

The equation is not used here to derive all gravitational dynamics. It is used to fix claim order. The left side is metric response in a faithful projection. The right side is a closed source ledger. Black-hole metrics are therefore read as licensed metric representatives of a source, vacuum, and boundary problem. When a model adds higher curvature, torsion, nonlocality, quantum expectation values, endpoint terms, or horizon-specific corrections, it changes tensor load or scope unless the added structure collapses to the same first-response form.

For black holes this has an immediate consequence. A high-curvature region cannot silently authorize a repair of exterior standing. If the endpoint of evaporation, the interior near a singularity, or the neighborhood of an inner horizon requires new physics, that new physics must be declared as high-curvature tensor load, effective-scope correction, quantum-gravity scope, leakage/radiation channel, boundary certificate, or non-sink quotient. The metric breakdown may motivate the change; it does not itself supply a same-scope reconstruction map.

This ordering preserves general relativity. Schwarzschild, Kerr, Reissner-Nordstrom, and Kerr-Newman remain legitimate representatives in their domains [1-5,27,28]. What changes is the reading of their interiors. A metric extension supplies representation. It does not automatically supply exteriorly referenceable standing-bearing facts. The horizon theorem tests exactly that overread.

AASC does not begin with gravity as primitive force, curvature, metric field substance, or quantization target. Gravity is the constraint role required for the admissible physical interior: standing surface, observable quotient, admissible redescription, first-class closure, and multiplier-type presentation. Once that role is faithfully projected into local Lorentzian metric language, the minimal first-response same-scope field-equation form is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{closed}}.$$

This equation is not used here as a primitive premise. It is the minimal metric projection of the AASC gravity role under closed source continuation. The source tensor is not primitive mass-energy substance; it is the metric-facing ledger of field-carrier, particle persistence, deformation response, bounded transport, interaction, and vacuum-baseline roles. The cosmological term is not primitive zero-point vacuum energy; it is a quotient-stable homogeneous metric-facing ledger.

The black-hole manuscript uses this result as a floor. Classical black-hole geometries are evaluated inside the licensed metric projection. That preserves the legitimacy of Schwarzschild, Reissner-Nordstrom, Kerr, and Kerr-Newman representatives while blocking metric backflow. A metric feature cannot authorize the admissibility architecture that made the metric meaningful.

3.2 10. Classical black-hole representatives and role separation

The standard classical representatives can be read as role packages [1-5,27,28].

The Schwarzschild exterior isolates the vacuum mass response of a non-rotating, uncharged, stationary black-hole exterior. Its mass parameter is exterior gravitational response. Its horizon is a metric/causal structure. Its central singularity is a classical metric-exhaustion diagnostic.

The Reissner-Nordstrom exterior adds electromagnetic charge. Its charge parameter is not a gauge-potential convention; it is fixed by the electromagnetic exterior response under declared gauge-boundary conditions. Its inner horizon raises separate continuation questions.

The Kerr exterior adds angular momentum. Its off-diagonal metric behavior and frame-dragging response are exterior rotational data. The black-hole spin

parameter is not a microscopic spin ontology; it is angular momentum in a stationary axisymmetric exterior role.

The Kerr-Newman exterior combines all three. It is therefore the natural classical classification target for this manuscript. It shows why the tuple (M, q_{BH}, J) is physically powerful while still being exterior role-data rather than interior fact inventory.

This role separation prevents a common slide. A classical metric may be exact in its own envelope. Exactness of the representative does not imply that every interior coordinate construction has the same standing as exterior observables. A complete classical solution is not the same as a complete exterior reconstruction certificate.

A classical black-hole solution is a metric representative with a declared envelope. In Schwarzschild form the exterior mass parameter organizes gravitational response. In Reissner-Nordstrom form an electromagnetic charge ledger is added. In Kerr form angular momentum and rotational response enter. In Kerr-Newman form all three exterior response roles are present. These metrics may be exact classical solutions, useful approximations, or idealized limits. Their AASC status depends on the role each component performs.

The metric representative is not the black-hole object by itself. The admissible black-hole object includes the exterior witness family, parameter ledger, horizon boundary typing, thermodynamic ledger, radiation or leakage profile, and continuation/certificate status. A metric chart can display an interior. It cannot by itself fix interior facthood. A maximal extension can be mathematically coherent. It does not automatically supply exterior standing for every region in the extension.

High-curvature or endpoint modifications are typed, not dismissed. Higher-curvature corrections are declared higher-response tensor load. Nonlocal terms expand the continuation envelope. Torsion or metric-affine nonmetricity changes connection/comparison anchors. Semiclassical sources require mixed projection and record closure. Horizon endpoint claims require certificate, leakage, non-sink quotient, or richer scope. If none is declared, the proposed term is hidden repair.

3.3 11. Metric regime theorem

Theorem 1 - Metric regime theorem. A black-hole metric regime is admissible as a same-scope physical description only when its metric descriptors are internal to a licensed projection, its exterior witnesses are declared, its parameter ledgers are role-typed, its horizon structures are boundary-typed, and its continuation relation preserves standing or classifies exhaustion.

Proof. The metric projection supplies comparison and curvature response but not primitive ontology. Exterior witnesses are required because standing-bearing reference must be re-fixable under admissible redescription. Parameter ledgers are required because mass, charge, and angular momentum do not stand as

bare labels. Boundary typing is required because a horizon label alone carries no reconstruction force. Continuation is required because black-hole claims routinely concern formation, evaporation, stability, radiation, and information recovery. If any of these fields is absent, the claim either changes scope, collapses to skin, imports hidden load, or fails closed. Hence the black-hole metric regime is the role package, not the metric representative alone.

4 Part IV - The Black-Hole Reconstruction Object

4.1 12. Fixed exterior black-hole scope

The fixed-scope requirement is the main protection against equivocation. In ordinary black-hole discussion one can move between asymptotically flat collapse, eternal two-sided geometries, AdS black holes, evaporating semiclassical spacetimes, algebraic exterior regions, stretched horizons, and boundary duals. Each of these may be useful. They are not the same theorem object. A claim that succeeds in one may not be a same-scope solution of another.

For example, an AdS/CFT boundary reconstruction claim may be an excellent certificate in a boundary-defined scope. It does not automatically solve an asymptotically flat exterior reconstruction problem unless a map between scopes is declared. A two-sided eternal black hole may represent thermofield-double structure. It does not automatically describe one-sided collapse and evaporation. A final-state projection may be a coherent model if declared as an endpoint boundary condition. It is not a same-scope continuation of a model that did not contain that boundary condition.

The tuple S_{BH} therefore acts as a guardrail. It forces each black-hole claim to say what the exterior domain is, what the horizon boundary is, which exterior references must be preserved, which witnesses count, which continuation relation is used, which physical assumptions are envelope data, which parameter and thermodynamic ledgers are in scope, and which certificates are available.

Define the fixed exterior black-hole reconstruction scope as

$$S_{\text{BH}} = (E_{\text{ext}}, BH, R_{\text{ext}}, W_{\text{ext}}, Cont, Env, Param, Therm, Cert).$$

Here E_{ext} is the exterior admissible description domain; BH is the candidate horizon or horizon-like boundary; R_{ext} is the family of standing-bearing exterior reference classes; W_{ext} is the exterior witness family; $Cont$ is the continuation relation; Env records physical assumptions, approximations, and model envelope; $Param$ records mass, charge, and angular-momentum response data; $Therm$ records thermodynamic ledger content; and $Cert$ records standing-transport certificate status.

The black-hole regime object is

$$B = (S_{\text{BH}}, g, H, Q_{\text{ext}}, A_{\text{wit}}, T_H, \text{Cert}_H, \Theta_{\text{BH}}).$$

The metric projection is g ; the horizon structure is H ; the exterior standing quotient is Q_{ext} ; the exterior witness algebra is A_{wit} ; the horizon standing-transport operator is T_H ; the certificate family is Cert_H ; and Θ_{BH} is the role-classification map assigning each claim its black-hole normal-form status.

This definition is intentionally exterior-scope relative. It does not claim that the entire universe has only the exterior domain. It fixes the domain in which the black-hole reconstruction claim is being evaluated. A larger quantum-gravity theory, boundary theory, island prescription, or interior model may be legitimate; it becomes same-scope only if it factors back through the declared witness and standing data.

4.2 13. Exterior standing quotient

The quotient is not optional. Without it, black-hole reasoning becomes sensitive to representative choices that carry no physical standing. A horizon may be described in different coordinates, by different slicings, or by different extension diagrams. Radiation may be described in occupation-number language, algebraic language, boundary correlators, or code-subspace language. The exterior standing quotient keeps these descriptions from being mistaken for different physical standing claims when they preserve all witness, continuation, boundary-access, and admissibility-status profiles.

The quotient also prevents false equivalence. Two descriptions are not equivalent merely because they share a verbal slogan such as “the information is preserved.” If one description supplies a radiation reconstruction map and another supplies only global retention, their witness profiles differ. If one description changes the asymptotic boundary algebra, its scope differs. If one description introduces a final-state condition, its anchor structure differs. The quotient therefore cuts both ways: it removes skin and protects tensor content.

A useful test is this: if a proposed redescription changes no element of Φ_W , Φ_C , Φ_B , or Φ_A , then it does not change exterior standing. It may still be a clearer presentation, a more efficient computation, or a useful analogy. But it is not a new standing-preserving recovery route. If it changes one of the profiles, it must declare whether it is adding a witness, changing boundary access, altering continuation, or modifying admissibility status. This is the local discipline that later controls holography, islands, remnants, and final-state projection.

For exterior reference representatives $r, r' \in R_{\text{ext}}$, define four profiles:

$$\Phi_W(r) = \{w \in A_{\text{wit}} : w \text{ re-fixes or distinguishes } r\},$$

$$\Phi_C(r) = \{r' \in R_{\text{ext}} : r \text{ Cont } r' \text{ by admissible continuation}\},$$

$\Phi_B(r)$ = the boundary accessibility, leakage, or horizon-trace profile of r ,

and

$\Phi_A(r)$ = the admissibility, role-type, and transport-status profile of r .

Define

$$r \sim_{\text{ext}} r'$$

iff the four profiles agree. The exterior standing quotient is

$$Q_{\text{ext}} = R_{\text{ext}} / \sim_{\text{ext}} .$$

A same-scope black-hole reconstruction preserves exterior standing exactly when it preserves this quotient together with the induced witness, continuation, boundary, and admissibility-status profiles. This quotient removes irrelevant representatives while preserving every standing-bearing difference.

This construction is essential for black holes because many descriptions use different coordinates, slicings, boundary languages, Hilbert-space presentations, or dual dictionaries. Those differences are harmless if they preserve Q_{ext} . They are not harmless if they alter exterior witnesses, continuation, boundary access, or transport status. The quotient is therefore the object against which holography, complementarity, remnants, islands, and global-state claims must be tested.

4.3 14. Exterior witness algebra

The witness algebra is deliberately broad. It includes not only direct detector readouts but also lawful functionals of exterior fields, asymptotic charges, radiation correlations, scattering data, gravitational memory, boundary correlators, thermodynamic histories, and reconstruction algorithms that are themselves anchored in declared data. The algebraic notation records closure under admitted exterior operations. It does not assert that every such algebra is already constructed in a complete quantum-gravity theory.

This breadth is important because it avoids an overly operational reading. AASC is not saying that a standing class must be read off by a local laboratory apparatus at a finite time. It may be re-fixed asymptotically, by correlations, by boundary reconstruction, by an error-correcting code, by a gravitational memory

record, or by a quotient identification. The exclusion is not against sophisticated reconstruction. It is against unlicensed reconstruction.

The witness algebra also handles negative cases. Suppose a global state contains a label identifying the infalling matter, but no exterior functional, boundary map, code subspace, or leakage channel connects that label to the old exterior witness algebra. Then the label may be meaningful in a larger domain, but it is not an old-scope exterior witness. Suppose a remnant has a large internal state space, but no exterior map distinguishes which state corresponds to which old reference. Then the remnant has capacity but not exterior standing. Suppose a Page curve turns downward, but no correlation map identifies threatened reference classes. Then the curve is evidence about entropy behavior, not yet a standing certificate.

The exterior witness algebra is written schematically as

$$A_{\text{wit}} = \text{Alg}(W_{\text{ext}}, A_{\partial H}^{\text{adm}}, \text{Cont}, \text{Cert}_H).$$

The notation does not assert that a complete algebraic quantum-gravity observable algebra has been constructed. It records the declared closure of exterior records, boundary traces, asymptotic observables, radiation data, correlation functionals, and admissible reconstruction maps. A symbol that is not generated by this package, does not change the exterior witness profile, and does not supply a certificate has no same-scope standing force.

Typical elements of A_{wit} include detector records, asymptotic gravitational or electromagnetic data, radiation correlations, flux measurements, boundary correlators, horizon traces, ringdown profiles, thermodynamic records, and declared reconstruction maps. If a proposed witness changes what the exterior can observe or reconstruct, it changes the witness package. If it has no exterior coupling, it is disconnected from exterior reconstruction. If it chooses a reconstruction without exterior basis, it imports a selector. These alternatives exhaust the way a purported witness can acquire standing in the old scope.

4.4 15. Black-hole target-regime instantiation

Theorem 2 - Black-hole target-regime instantiation. Black-hole physics instantiates the AASC target class whenever it is used to make same-scope claims about persistent exterior objects, horizon histories, conserved parameters, thermodynamic ledgers, radiation, or information reconstruction.

Proof. Such claims require reference to a black-hole object or history; standing for exterior records and parameter ledgers; admissibility conditions distinguishing licensed reconstruction from arbitrary redescription; and irreversibility preventing failed transport from later being counted as the same successful transport. The target cannot be stated without stable exterior witnesses, metric comparison,

continuation relations, and role-typed boundary structures. Therefore the AASC kernel is instantiated by the black-hole reconstruction problem itself.

5 Part V - Threatened Exterior Standing and Horizon Isolation

5.1 16. Threatened exterior reference classes

The word “threatened” is scope-relative. It does not mean that every microscopic detail must be reconstructed in every black-hole model. It means that the model has declared a reference class whose standing is to be preserved, and that the class becomes dependent on horizon-isolated structure.

A coarse classical collapse model may declare only mass and angular momentum response. In that case most microscopic features of the infalling body are outside the declared burden. A quantum-information model may declare a much richer reference class: purification with an exterior system, distinguishability of infalling states, or reconstructability from Hawking radiation. A holographic model may declare a boundary algebra. A remnant model may declare a late storage carrier. Each target produces a different burden.

This is why the horizon theorem is not a demand for maximal information recovery. It is a demand for target honesty. If the target is only exterior parameters, parameter preservation may suffice. If the target is full state reconstruction, parameter preservation does not suffice. If the target is global unitarity, exterior reconstruction still requires a map from the global object to the exterior witness algebra.

The reference classes below recur throughout the paper:

- exterior preparation classes, such as an infalling probe prepared and recorded before horizon dependence;
- matter-source classes contributing to exterior mass, charge, and angular momentum;
- radiation-correlated classes claimed to be recoverable from outgoing quanta;
- entangled-reference classes involving an exterior purifier or algebraic reference;
- conserved parameter classes preserving compressed exterior response;
- perturbation and ringdown classes witnessing exterior metric response.

The horizon isolation lemma applies only to classes with declared standing. It does not invent a burden the model never claimed.

The examples also distinguish levels of preservation. A conserved exterior parameter may remain fixed even when the detailed reference class of the infalling object is not recovered. If a charged shell collapses, the exterior charge can remain visible through electromagnetic response. That preserves the charge role. It does not preserve all standing-bearing distinctions among possible shells

with the same charge. The same point holds for mass and angular momentum. Exterior parameters are powerful compression of response; they are not complete microstate records unless a model supplies the extra distinguishability.

A perturbation class gives a second illustration. Ringdown frequencies can witness the exterior parameter package and aspects of the horizon structure. They can distinguish a Kerr-like exterior from a Schwarzschild-like one, and they can constrain deviations. But ringdown by itself does not reconstruct every field configuration that fell in. It preserves standing at the level of the response role. Treating it as full interior fact recovery would be an overpromotion.

An entangled-reference class illustrates why information language must be typed. If an exterior mode is entangled with an infalling mode, one can write a global quantum state. But the standing question is not whether a symbol in the global state exists. It is whether the declared exterior algebra can re-fix the relevant reference after horizon isolation. An algebraic reconstruction, island prescription, or holographic code may do that. A bare statement that the global state is pure does not.

Thus the threatened-reference inventory is not meant to inflate the theorem target. It narrows it. The theorem applies only to reference classes that the declared exterior reconstruction problem requires to preserve. If a model does not claim to reconstruct a class, then AASC does not accuse the model of failing that reconstruction. If a model does claim preservation, it must say where the standing-bearing image appears.

The central proof must not begin by assuming that horizon transport is undefined. It begins with physical black-hole reference classes whose exterior standing is intelligible before horizon isolation and whose preservation becomes a declared reconstruction burden. A threatened exterior reference class is a class $r \in R_{\text{ext}}$ whose standing must remain re-fixable in the declared exterior reconstruction problem but whose continuation becomes dependent on a horizon-isolated carrier.

Several physically recognizable classes are relevant.

First, an **infalling-probe record class** is created when an exteriorly prepared probe, detector, matter packet, or field excitation has exterior standing before horizon crossing. The exterior problem may later ask whether the same probe-content is re-fixed by radiation, boundary data, or correlations. Without a certificate, the old exterior can no longer distinguish that standing-bearing content after permanent isolation.

Second, a **matter-source continuation class** arises when matter or field content contributes to mass, charge, angular momentum, or other exterior response before horizon isolation. Some of its role-data may remain exteriorly fixed through conserved parameters; the full standing-bearing content of the infalling matter is not thereby recovered. Parameter preservation is not arbitrary microstate preservation.

Third, a **radiation-correlated reference class** is a class whose later exterior

standing is claimed through correlations in Hawking radiation, gravitational radiation, electromagnetic radiation, or other outgoing channels. It has standing only if the radiation record is declared, encoded, and linked to the threatened reference by a redescription-stable map.

Fourth, an **entangled-reference class** is a class involving exterior quantum correlations. It is standing-bearing only relative to a declared subsystem, algebra, code, or reconstruction map. Entanglement entropy or Hilbert-space norm alone does not re-fix the class.

Fifth, a **conserved exterior parameter class** is fixed by mass, charge, or angular-momentum response. These classes are important because they show that some exterior standing may remain while other reference content is lost. If charge remains exteriorly fixed by electromagnetic flux, that does not certify every interior microstate.

Sixth, a **perturbation or ringdown class** is fixed by exterior response to perturbations. Ringdown, frame dragging, quasi-normal response, or scattering data can witness parameter and horizon structure. They may certify exterior role-data without certifying hidden interior facthood.

These examples show why R_{ext} is not a purely formal placeholder. It is the family of reference classes the black-hole reconstruction problem itself requires when it talks about formation histories, infalling systems, exterior records, radiation, conserved parameters, and recoverable information.

5.2 17. Horizon isolation

Horizon isolation has two components. The first is metric or causal: the horizon boundary prevents direct signal return or exterior access in the relevant representative. The second is reconstructive: no admitted exterior structure supplies an indirect image of the threatened reference class. The theorem requires both.

If Hawking radiation, boundary correlations, or exterior fields carry a standing image, then the horizon is not an uncertified sink. If a holographic boundary map reconstructs the class, the certificate exists. If complementarity supplies a redescription-stable correspondence, the model exits the uncertified case. If a remnant exposes an exterior map, it may function as a non-sink endpoint. The isolation lemma targets the remaining case: permanent horizon dependence plus no old-scope image.

The lemma also clarifies why an inaccessible interior is not automatically a contradiction. A larger theory may contain interior structure. The old exterior problem fails only when the model also claims exterior standing preservation without supplying an exterior image. The theorem is therefore reconstructive, not global-ontological.

A horizon isolates exterior standing relative to a scope only when the old exterior witness package supplies no future image for the threatened class. Permanent

horizon isolation is therefore a relation among a horizon boundary, a reference class, and a witness algebra. It is not simply the existence of a classical event horizon in a metric solution.

Let E_{int} denote the carrier envelope that is permanently isolated relative to the exterior scope. A reference class becomes horizon-dependent when its continuation requires access to E_{int} or to a certificate representing E_{int} in exterior data. If the model supplies leakage, radiation, a boundary trace, holographic reconstruction, complementarity correspondence, remnant map, or non-sink quotient, then the class may still have exterior standing. If none is supplied, the class lacks old-scope exterior re-fixability.

This is the black-hole-specific physical bridge to the partial-domain theorem. The operator T_H is not partial by stipulation. It becomes partial because the old exterior scope lacks an admissible witness image for the threatened class.

5.3 18. Horizon Isolation Lemma

The lemma is the hinge of the manuscript. It converts black-hole horizon structure into the partial-domain obstruction without treating partiality as a definition. The inputs are physical and scope-relative: a reference class is standing-bearing, the exterior problem requires re-fixability, the class becomes dependent on the isolated carrier, the horizon is permanent relative to the old exterior, and no recognized route back to the exterior witness algebra is supplied.

The no-certificate hypothesis is also structured. It excludes ordinary leakage, Hawking-like radiation used as a witness, explicit boundary traces, holographic maps, complementarity correspondences, remnant maps, and non-sink quotient structures. It does not exclude those routes globally. It says that if one of them is supplied, the case is no longer the uncertified permanent case. The theorem is therefore not hostile to physical solutions. It separates solutions from assertions.

One may object that the old exterior witness algebra could be incomplete. That is correct, but it is not a same-scope counterexample. If additional witnesses are admitted, the scope has changed or a certificate has been supplied. The old theorem object is no longer being preserved. This is why the horizon theorem is scope-locked. It classifies same-scope exterior reconstruction, not all possible future theoretical enlargements.

A second objection is that a sufficiently clever reconstruction algorithm might recover the class without changing the witness algebra. If the algorithm is a functional of the old exterior witnesses, then it belongs to A_{wit} and the case is certified. If it uses data not in the old algebra, the scope has changed. If it chooses among possibilities not fixed by the old data, it is a selector. The algorithmic route is therefore included rather than ignored.

Theorem 3 - Horizon Isolation Lemma. Let S_{BH} be a fixed exterior black-hole reconstruction scope with witness algebra A_{wit} and exterior standing quotient Q_{ext} . Let H be a candidate horizon boundary. Suppose:

1. $r \in R_{\text{ext}}$ is a standing-bearing exterior reference class;
2. the declared reconstruction problem requires r to remain re-fixable under admissible exterior continuation;
3. the continuation of r becomes dependent on E_{int} , the horizon-isolated carrier;
4. H is permanent relative to S_{BH} ;
5. no leakage channel, exterior boundary trace, radiation witness, holographic anchor map, complementarity correspondence, remnant map, or non-sink quotient is supplied.

Then r has no admissible image in the old exterior witness algebra A_{wit} .

Proof. Standing for r requires re-fixability under the declared witness algebra, continuation relation, or certificate package. By hypothesis, the reconstruction problem requires such re-fixability; otherwise r would not be part of the standing-bearing exterior burden. By hypothesis, the continuation of r becomes dependent on the horizon-isolated carrier E_{int} . Because the horizon is permanent relative to the old exterior scope, direct exterior access to that carrier is not available as an ordinary future exterior witness. Because no leakage or radiation channel is supplied, there is no outgoing record carrying the reference into A_{wit} . Because no boundary trace, holographic anchor map, complementarity correspondence, remnant map, or non-sink quotient is supplied, there is no admissible replacement route by which the old exterior witness algebra re-fixes r .

Assume nevertheless that r has an admissible old-scope exterior image. If the image is generated by A_{wit} , then the relevant witness, trace, correlation, or reconstruction map was already present, contradicting the no-certificate hypothesis. If the image is generated by new data, the witness algebra has changed and the proposal is larger-scope or certified. If the image is chosen without witness support, it is selector import. If it is added only after transport failure, it is repair. If it changes no exterior standing profile, it is bookkeeping. These cases exhaust the old-scope acquisition routes. Therefore r has no admissible image in the old exterior witness algebra.

5.3.1 Physical example: infalling probe or reference qubit

The abstract notation can be read in the standard black-hole information example. Let an exterior laboratory prepare a reference-bearing system r - for instance a probe, field excitation, or reference qubit entangled with a retained exterior register - and record its preparation before the system becomes horizon-dependent. At that stage the exterior scope has standing for the preparation class: the laboratory record, exterior register, or preparation apparatus fixes what is meant by the reference.

Now suppose the system crosses a candidate horizon and the model later claims that the same reference is preserved. There are several admissible ways this can happen. Hawking radiation or other leakage may carry correlations that re-fix the reference. A holographic boundary theory may reconstruct the corresponding

operator. An island prescription may identify a radiation algebra that includes the relevant reconstruction. A remnant may expose a late exterior map distinguishing the stored class. Complementarity may supply a stable correspondence. Each of these supplies, or attempts to supply, a certificate.

If none of those routes is declared, the old exterior cannot re-identify the reference class. The statement that the qubit, probe, or field excitation “still exists behind the horizon” may be meaningful in a larger representation, but it is not an old-scope exterior witness. A later assignment pairing an outgoing symbol with the old reference would choose a representative not fixed by the old exterior algebra; that is selector import. If the pairing is introduced only after the loss of exterior re-fixability, it is repair. If a new boundary algebra or quantum-gravity sector is added, the scope has changed or a certificate has been supplied. The infalling-probe example is therefore the physical counterpart of the partial-domain theorem: uncertified permanence makes the exterior standing-transport map undefined on the horizon-isolated carrier.

5.4 19. Partial-domain induction

Lemma 4 - Partial-domain induction. If a threatened exterior reference class r has no admissible image in A_{wit} , Cont , leakage, certificate, or non-sink quotient, then the horizon standing-transport operator is undefined on the corresponding old-scope horizon-isolated carrier.

Proof. A standing-transport operator assigns exterior re-fixability data to carrier elements. If an element of the horizon-isolated carrier supports or contains the threatened continuation of r , then defining T_H on that carrier would supply an exterior image of r . By hypothesis no such image exists in the old witness algebra, continuation relation, leakage channels, certificates, or quotient structure. Thus T_H is not merely unknown at that carrier; it is not in the admissible domain of the operator. Undefinedness records non-membership in the old-scope transport domain.

6 Part VI - Horizon Partial-Domain Obstruction

6.1 20. Horizon carrier decomposition

Let the source-horizon-exterior carrier be

$$X_H = E_{\text{ext}} \cup E_{\partial H} \cup E_{\text{int}}.$$

Here E_{ext} is the exterior witness envelope; $E_{\partial H}$ is the boundary or horizon trace envelope; and E_{int} is the interior or permanently isolated envelope relative to the exterior scope. The horizon standing-transport assignment is a partial operator

$$T_H : X_H \rightharpoonup R_{\text{ext}}$$

with domain

$$\text{dom}(T_H) = E_{\text{ext}} \cup E_{\partial H}^{\text{cert}}.$$

The certified boundary component $E_{\partial H}^{\text{cert}}$ contains boundary traces, leakage data, radiation records, holographic maps, complementarity correspondences, remnant maps, or quotient data that actually transport standing. In the uncertified permanent case,

$$E_{\text{int}} \cap \text{dom}(T_H) = \emptyset.$$

This is a theorem-induced partiality. The previous section established that threatened classes lack an old-scope exterior image. The partial operator records that absence.

6.2 21. No-totalization theorem

The no-totalization theorem is the mathematical heart of the paper. It can be read as a fixed-domain partial algebra result. The old exterior problem has a partial standing-transport operator. Its domain is fixed by the old witnesses and certificates. An old undefined application cannot become defined while all old anchors and tensors are preserved. If it becomes defined, something did the defining. That something is either already in the old scope, in which case the application was not undefined, or it is new structure, in which case the scope or certificate status has changed.

This is the precise sense in which a permanent uncertified horizon fails. The failure is not that the word “horizon” is forbidden. The failure is that the model simultaneously asserts horizon-isolated threatened standing, denies every admissible image, and then treats the reference as preserved. Those three commitments cannot coexist in a same-scope reconstruction object.

The theorem also separates evaporation from proof assumption. If evaporation provides leakage, it satisfies the structural disjunction. If a model instead uses a holographic certificate or non-sink quotient, it may also satisfy the disjunction. If no continuing successor exists, exhaustion is allowed. What is not allowed is uncertified permanence as reconstruction.

The theorem can be read as a no-repair result for old-scope operator domains. The domain of T_H is not a matter of ignorance. It records which carrier elements have an admissible standing-transport assignment in the declared exterior scope. When the Horizon Isolation Lemma applies, the relevant carrier elements are outside the domain. A totalization that makes them defined has changed the standing status of the old carrier.

This is different from learning an unknown value. If an exterior observer has not yet decoded radiation but the radiation already carries a declared certificate,

the operator may be defined even if the value is practically inaccessible. If no such certificate exists, the undefinedness is structural. It cannot be removed by saying that the information is somewhere in the global state.

The source of totalization authority must be named. If it is a boundary dictionary, the dictionary is a certificate only when it preserves the exterior standing quotient. If it is a remnant label, the label is a certificate only when the exterior can map it to threatened classes. If it is a final-state condition, the condition is an anchor only when declared as physical data. If it is a Page-curve inference, the inference is certificate-relevant only when it identifies the reference classes, not merely an entropy trend.

This is why the no-totalization theorem is robust across interpretive programs. It does not care whether the proposed recovery is semiclassical, holographic, algebraic, stringy, effective, or quantum-informational. It asks the same fixed-domain question: what makes the old undefined transport defined without changing scope, selecting a representative, or repairing after failure?

Theorem 5 - No same-scope horizon totalization. No same-scope operator

$$T_H^* : X_H \rightarrow R_{\text{ext}}$$

can extend T_H , be defined on E_{int} , and preserve the old exterior standing quotient Q_{ext} , witness algebra A_{wit} , continuation relation $Cont$, parameter ledger $Param$, thermodynamic ledger $Therm$, and certificate status, unless a standing-transport certificate, leakage channel, non-sink quotient, or scope change is supplied.

Proof. On E_{ext} and certified boundary data, T_H is defined by exterior witnesses, continuation, or certificates. On the uncertified permanent interior envelope, the Horizon Isolation Lemma and Partial-Domain Induction Lemma show that T_H is undefined. A totalization T_H^* would assign an exterior reference image to old-scope carrier elements whose standing transport is not defined.

If the assignment is determined by the old witness algebra, then T_H was not undefined and the case is certified. If the assignment is not determined by the old witness algebra, a representative or continuation has been selected without a witness; this is hidden selector import. If new exterior data are added, the witness algebra changes or a certificate is supplied. If an undefined old-scope assignment becomes defined only after failure, the move is repair. If a derived item such as entropy, global norm, dual calculability, remnant capacity, or algorithmic success is treated as primitive authorization, the move launders a carrier. If no standing-bearing witness changes, the totalization is bookkeeping and does not preserve the threatened class. These alternatives exhaust the possible acquisition of an exterior image. Therefore no same-scope totalization exists without certificate, leakage, non-sink quotient, or scope change.

6.3 22. Finite witness model for horizon isolation

The finite witness model is deliberately small. It does not model the full field theory of a black hole. It isolates the logic of standing transport.

The exterior points e_1, e_2 stand for ordinary exterior witness-bearing configurations. The boundary point b stands for a horizon trace that is actually declared in the exterior scope. The hidden points h_1, h_2 stand for horizon-isolated carriers. The partial operator records exactly where the old exterior problem has an image. If a certificate is later supplied for h_i , then h_i moves into $E_{\partial H}^{\text{cert}}$ or an equivalent certified domain. The theorem does not forbid that move. It forbids pretending that the move was already old-scope transport.

This toy model is useful because every attempted rescue is visible. A hidden choice function σ is not physics. A new witness is a certificate or scope change. A post-loss assignment is repair. A symbol that changes nothing is bookkeeping. The same alternatives appear in the full black-hole problem under more sophisticated names: global unitarity, holographic dictionaries, Page-curve inference, remnant labels, final-state maps, baby-universe branches, or quantum-gravity promises.

The finite model can be used in three variants. In the **certified variant**, add a boundary witness c_i with $T_H(c_i) = r_i$ and declare $c_i \in E_{\partial H}^{\text{cert}}$. The totalization is no longer illicit because the relevant carrier has entered the certified boundary domain. The horizon is not an uncertified standing sink.

In the **scope-changing variant**, add a new witness w'_i not generated by the old W_{ext} . Then the hidden point can be assigned an exterior image relative to the new algebra, but the old theorem object has changed. This may be good physics. It is not same-scope rescue.

In the **bookkeeping variant**, add a symbol s_i labeling h_i while all old witnesses remain unchanged. If s_i has no exterior effect, it is skin. If it is treated as standing-bearing despite no exterior effect, it is hidden load. Thus labels alone do not preserve exterior standing.

These variants show that the obstruction is not anti-reconstruction. Certified reconstruction is allowed. Enlarged reconstruction is allowed when declared. What fails is untyped recovery: a claim that the old exterior reconstruction object is preserved while the standing-transport domain has silently changed.

A finite model makes the obstruction explicit. Let

$$X = \{e_1, e_2, b, h_1, h_2\},$$

where e_1, e_2 are exterior carriers, b is a boundary carrier, and h_1, h_2 are horizon-isolated carriers. Let

$$R_{\text{ext}} = \{r_1, r_2, r_b\}, \quad W_{\text{ext}} = \{w_1, w_2, w_b\}.$$

Define the partial transport operator by

$$T_H(e_i) = r_i, \quad T_H(b) = r_b, \quad T_H(h_i) \uparrow,$$

where $\uparrow$ denotes non-membership in the operator domain. An attempted totalization assigns

$$T_H^*(h_i) = r_{\sigma(i)}$$

for some choice function σ .

If σ is fixed by the old witness algebra, then the hidden carrier already had an exterior witness image and $T_H(h_i)$ was not undefined. If σ is not fixed by the old witness algebra, it is a hidden selector. If a new witness fixes σ , the witness package has changed or a certificate has been supplied. If the assignment is added only after horizon isolation has removed exterior re-fixability, it is repair. If no assignment is made, the exterior successor is exhausted. If a symbol is added without changing the witness profile, it is bookkeeping. Thus the finite witness model exhibits exactly the route structure of the general theorem.

6.4 23. No uncertified permanent-horizon reconstruction

Theorem 6 - No uncertified permanent horizon as continuing exterior reconstruction. Let S_{BH} be a fixed exterior black-hole reconstruction scope with nonempty standing-bearing exterior burden. Suppose a candidate horizon is permanent relative to S_{BH} and at least one threatened exterior reference class lacks leakage, exterior witness, boundary trace, standing-transport certificate, remnant map, holographic anchor map, complementarity correspondence, or non-sink quotient. Then the horizon is not an admissible continuing same-scope standing-preserving exterior reconstruction object.

Proof. The Horizon Isolation Lemma gives that the threatened class has no admissible image in the old exterior witness algebra. The Partial-Domain Induction Lemma gives that the horizon standing-transport operator is undefined on the corresponding old-scope horizon-isolated carrier. The No-Totalization Theorem gives that no same-scope totalization exists preserving the old exterior quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status. Therefore the model cannot claim continuing same-scope standing-preserving horizon reconstruction. If it supplies leakage, certificate, or non-sink quotient, it exits the uncertified case. If it changes scope, it is a larger-domain proposal. If no successor exists, the correct classification is exterior-reconstruction exhaustion. The excluded claim is uncertified permanence as continuing same-scope reconstruction.

6.5 24. Successor exhaustion and counterexample burden

Define the exterior admissible successor set:

$$Succ_{S_{BH}}(H^\partial) = \{K^\partial : H^\partial \rightsquigarrow K^\partial \text{ by standing-preserving admissible exterior continuation}\}.$$

If

$$Succ_{S_{BH}}(H^\partial) = \emptyset,$$

then the exterior reconstruction problem has exhausted in the declared scope. This is not evaporation, not thermodynamic depletion, not proof of global nonexistence, and not constraint failure. It is absence of a standing-preserving same-scope successor. A model may acknowledge exhaustion. It may not rename exhaustion as successful permanent-horizon continuation.

A counterexample to the horizon theorem must supply a fixed S_{BH} , a nonempty standing-bearing exterior burden, a permanent horizon, no leakage, no evaporation, no certificate, no non-sink quotient, preserved Q_{ext} , preserved A_{wit} , preserved $Cont$, preserved $Param$, preserved $Therm$, a totalized T_H^* , no selector, no repair, no carrier laundering, no scope change, no bookkeeping equivalence, and no exterior-successor exhaustion. The no-totalization theorem states that such an object cannot be constructed inside the declared scope.

7 Part VII - Singularities and High-Curvature Endpoints

7.1 25. Failure-signature frame

The singularity analysis uses the same fixed-domain method as the horizon analysis, but the target is different. A singularity is not primarily a standing-transport boundary. It is a diagnostic of a representation that cannot be extended while preserving the relevant metric commitments. Horizons can be nonsingular in curvature but still obstruct reconstruction. Singularities can mark curvature or continuation failure without functioning as horizons. Confusing these roles makes black-hole discussions unstable.

AASC therefore refuses two common overreads. The first overread treats the singularity as a physical object that stores, destroys, or emits standing. The second treats the singularity as an admissibility boundary that can terminate the reconstruction problem. Both uses require more than the diagnostic supplies. If a singularity is to do physical work, that work must be represented by declared tensor load, endpoint dynamics, a boundary condition, or a richer-scope model.

Singularity language enters only after a metric projection is licensed. The governing order is

$$Governance \rightarrow Admissibility \rightarrow Representation \rightarrow Diagnostics.$$

A diagnostic cannot ground governance. A metric singularity indicates that a fixed representational commitment has exceeded its admissible envelope. It is not a primitive object waiting at the bottom of the black hole.

The black-hole singularity theorem is therefore not a resolution of singularities. It is a classification. Singular behavior may signal curvature blow-up, geodesic incompleteness, extension failure, Cauchy-horizon instability, or high-curvature endpoint transition. None of these by itself supplies a terminal standing-bearing fact or an admissibility boundary.

7.2 26. Schwarzschild and Kerr cases

The Schwarzschild case supplies the cleanest contrast. The horizon at $r = 2M$ is not a curvature singularity; the representative can be extended across it. The central singularity at $r = 0$ is a curvature and geodesic-incompleteness diagnostic. The horizon theorem concerns the former only when exterior standing preservation is claimed. The singularity theorem concerns the latter as metric exhaustion. Neither role authorizes interior facthood.

The Kerr case adds rotation and an inner structure. The ring singularity is not a primitive rotating object. It is a pathology of the classical metric extension. The inner horizon and associated Cauchy-horizon issues are continuation and stability problems. A model that uses Kerr interior structure for reconstruction must declare its continuation scope and witnesses. Otherwise it is using analytic extension as hidden authority.

These examples show why singularity classification is not anti-GR. It allows the classical representatives to do their mathematical work while blocking their promotion into ontic endpoints.

The Schwarzschild case also illustrates why horizon and singularity must be separated. The horizon is not a curvature singularity. It is a metric and causal boundary structure that depends on the global or representative envelope. The central singularity is a different diagnostic: it marks failure of the classical metric extension. AASC does not collapse these into one boundary. The horizon generates reconstruction questions; the singularity diagnoses metric exhaustion.

The Kerr case adds two complications. First, rotation creates an ergoregion and frame-dragging response exterior to the horizon. These are exterior rotational-response structures, not proofs of an interior rotating substance. Second, the analytic extension contains structures that are mathematically striking but scope-sensitive. Treating the ring singularity, inner horizons, or extended regions

as physical ontology requires declared witnesses and continuation ledgers. The metric alone does not provide those ledgers.

The Reissner-Nordstrom case makes the same point for charge. The charge parameter is exteriorly visible through electromagnetic response. The inner horizon and maximal extension raise continuation issues. But exterior charge standing does not certify the complete interior state. It preserves a gauge-boundary response role. It does not solve the information problem.

These examples locate the singularity theorem in ordinary black-hole physics. The theorem is not a generic slogan that singularities are unimportant. It says that singularities are important diagnostics, but diagnostics do not become primitive boundary objects or recovery mechanisms.

In the Schwarzschild representative, $r = 0$ is not treated as a physical point-object. It is the location at which the classical metric representation exhibits curvature pathology and geodesic incompleteness. AASC classifies this as metric-extension exhaustion. The horizon at $r = 2M$ is a different issue: it is a horizon boundary in the metric representation and becomes reconstruction-relevant only through exterior witness and certificate discipline.

In the Kerr representative, the ring singularity is likewise not an interior ontology generator. It is a metric-extension pathology in a rotating solution. If a model uses analytic extensions, chronology-violating regions, or alternative interior structures, it has declared a richer metric or global structure. Such a structure may be mathematically studied, but it does not automatically create same-scope exterior standing for hidden interior facts.

In the present fixed exterior reconstruction theorem, the Kerr ring singularity is treated as a metric-extension pathology or exhaustion signature. Claims about analytic extension, chronology violation, or alternative interior continuation belong to richer-scope geometry and require their own anchors, continuation ledgers, and witnesses.

In Reissner-Nordstrom and Kerr interiors, inner or Cauchy horizons require separate classification. They are continuation and determinacy structures, not simply ordinary singularity labels. If mass inflation, instability, or high-curvature development is invoked, that is high-curvature or effective tensor load. If a model claims stable continuation through such a region, it must supply the appropriate continuation, witness, and certificate ledgers.

7.3 27. Evaporating endpoints

An evaporating endpoint is not a free repair site. If semiclassical evaporation leads to a high-curvature terminal regime, the endpoint must be classified as one of the following: declared higher-response tensor load, lawful radiation or leakage channel, endpoint certificate, non-sink quotient, richer quantum-gravity scope, or exterior-reconstruction exhaustion. It cannot be used as an untyped location where information is restored, destroyed, or transported by fiat.

This matters for remnants, final-state projection, and baby-universe routes. A Planck-scale remnant cannot preserve exterior standing by capacity alone. A final-state condition cannot repair transport after the fact unless declared as physical boundary data in the original scope. A baby universe cannot preserve old exterior standing unless it induces a map back to the old exterior standing quotient.

7.4 28. Singularity classification theorem

Theorem 7 - Black-hole singularity classification. In a licensed black-hole metric regime, singularities and high-curvature endpoint structures are diagnostics of exhaustion or transition of the metric representation. They do not by themselves supply physical objects, standing-bearing terminal facts, horizon certificates, or admissibility boundaries.

Proof. Singularities appear inside metric representation. Their mathematical forms may include curvature divergence, incomplete curves, failed extension, or instability of continuation. Each form presupposes the metric projection and its representational envelope. A diagnostic of failure within that envelope cannot authorize the boundary conditions that licensed the envelope. If a singular or high-curvature structure is used to carry standing-bearing physical content, the model must declare the carrier, witness, continuation, and certificate ledgers. Without those, the singularity is a diagnostic, not an ontology. Therefore black-hole singularities have exhaustion-signature status unless richer structure is supplied.

8 Part VIII - Mass, Charge, and Angular Momentum as Exterior Response Roles

8.1 29. Parameter discipline

A black-hole exterior standing quotient is role-typed before it is read as ontology. In the familiar settled Kerr–Newman example, the exterior parameter quotient is compressed by mass, electric charge, and angular momentum. AASC preserves this compression as exterior role-data. It rejects only the promotion of those exterior descriptors into hidden interior facthood or into a certificate for finer standing-bearing distinctions.

The exterior parameter package is written

$$P_{\text{BH}} = (M_{\text{ext}}, q_{\text{BH}}, J_{\text{ext}}).$$

The notation avoids using Q_{ext} for charge, since Q_{ext} already denotes an exterior standing quotient. The charge parameter is written q_{BH} .

The classical three-parameter form should be read as exterior quotient compression. It does not prove that no larger microphysical structure, hair-like witness, or richer-scope description can be declared. It says that the declared exterior response quotient re-fixes only the content represented by its admitted parameters and witnesses. That is a powerful physical fact. It is not a license to treat hidden interior labels as standing-bearing unless a model supplies exterior distinguishability or reconstruction.

8.2 Exterior quotient, hair, and metric regime

The proof-level black-hole object is the declared exterior standing quotient, not stationarity and not exterior-quotient as a metaphysical premise. In the familiar settled Kerr–Newman example this quotient is represented by mass, electric charge, and angular momentum,

$$Q_{\text{ext,par}}^{\text{BH}} = (M_{\text{ext}}, q_{\text{BH}}, J_{\text{ext}}),$$

but this is a clean standard exterior-parameter example rather than a restriction on the theorem. A dynamical collapse spacetime, evaporating semiclassical model, ringdown regime, perturbed horizon, quasi-local horizon framework, or richer quantum-gravity description may add exterior witnesses, radiation records, perturbative correlations, boundary data, or certificate maps. Those additions subtract from the unresolved residue precisely to the extent that they re-fix standing-bearing content. If they do not re-fix standing, they are skin, bookkeeping, hidden load, repair, selector import, carrier laundering, or scope change.

Hair-language is therefore role-typed rather than treated as an escape category. If alleged hair is exteriorly measurable and changes the declared standing quotient, it is exterior tensor or witness content. If it supplies a boundary, holographic, island, radiation, remnant, or quotient reconstruction map, it is a certificate. If it is a hidden label with no exterior witness effect, it is bookkeeping or hidden load. If it adds a new sector, algebra, or quantum-gravity structure, it is richer scope. If it pairs old standing with a new representative only after standing loss, it is repair or selector import. There is no fourth same-scope role called hair that evades anchor/tensor/skin discipline.

Thus the endpoint arc begins only after exterior-quotient subtraction: the declared exterior quotient, including any genuine hair-like witnesses and all certificates, removes exactly what it re-fixes. Any finer standing-bearing burden not re-fixed by that quotient remains in the unresolved residue.

8.3 30. Mass as exterior source response

The exterior mass role is also the place where the paper interfaces with gravitational radiation. In an evaporating or radiating setting, mass is not a static label. It may be tracked by a flux law, Bondi-type exterior ledger, or semiclassical

expectation value. Such changes are admissible when the flux or exchange channel is declared. They are not violations of standing. They are part of the continuation profile of the mass response.

This matters for information. A mass ledger can preserve the total energy response of what fell in, but it does not preserve the detailed identity of what fell in. The compression is lawful when the target is mass. It is insufficient when the target is microscopic state reconstruction. Thus exterior parameter sufficiency is target-relative. It suffices for the declared exterior parameter quotient; it does not suffice for every information target.

The distinction between mass roles also prevents confusion about evaporation. In an evaporating exterior, a Bondi-type mass ledger may change because radiation carries energy away. This change does not mean that the black hole's old mass was a primitive substance leaking out. It means the exterior source-response ledger has a flux term in the declared radiative scope. The mass role is transported through a ledger, not preserved as an untyped scalar essence.

Similarly, an irreducible mass derived from horizon area belongs to the thermodynamic and horizon ledger. It may be central for classical black-hole mechanics. It is not identical to ADM mass in every scope, and it is not a hidden interior payload. It is a role in a particular exterior/horizon comparison. The same variable may appear in equations, but AASC distinguishes the role played by the quantity.

This role separation is useful for remnants. A remnant with a small exterior mass but enormous alleged internal storage has not preserved exterior standing merely by being massive. The mass ledger records exterior gravitational response. The storage claim requires a separate reconstruction map. Collapsing the two is carrier laundering: using one role to perform another's work.

Black-hole mass is not a primitive scalar attached to an object. In the AASC mass/inertia source, mass first enters as an operative deformation-response profile of localized persistence identity, with inertia as the transport-state restriction. In a black-hole exterior, that role is represented by gravitational/source response and by the metric-facing source ledger.

Several standard mass notions illustrate the role discipline. ADM mass is an asymptotic exterior mass in an asymptotically flat scope. Bondi mass is a radiative exterior mass ledger in a flux-carrying scope. Komar mass is conditioned by stationarity and symmetry. Irreducible mass belongs to the horizon-area and thermodynamic ledger. These notions are not interchangeable primitive scalars; each has a scope and witness structure.

Theorem 8 - Black-hole mass role. A black-hole mass parameter is admissible in a fixed exterior black-hole scope only as an exterior gravitational, source-response, thermodynamic, or radiative class fixed by the declared witness ledger.

Proof. If the mass value is fixed by asymptotic gravitational response, it is an exterior response role. If it changes by radiation flux, the relevant mass ledger

is radiative. If it appears in the first law, it is a thermodynamic role. If it is introduced as a bare scalar with no witness, source, or response profile, it is skin or hidden load. Thus mass stands only through the appropriate exterior response ledger.

A hidden mass reservoir inaccessible to all exterior witnesses, deformation response, metric response, thermodynamic role, and continuation relation has no realized mass role in the declared exterior scope. If it becomes accessible, it belongs to the admitted profile. If it remains inaccessible, it is inert relative to exterior reconstruction.

8.4 31. Charge as gauge-boundary response

Charge illustrates how a hidden interior property can be unnecessary. Exterior electromagnetic flux can fix net charge without accessing interior constituents. Gauge presentation may vary, but the boundary response remains. This is exactly the kind of exterior standing AASC admits. The exterior charge class can survive horizon isolation because it has a boundary response. But this success is limited: net charge is a compressed invariant. It does not certify every charged degree of freedom that contributed to it.

This limitation matters for remnants and information. A remnant carrying charge may be exteriorly distinguishable by charge. That does not mean it exteriorly reveals all stored data. A holographic or radiation channel must supply the additional certificate if the target requires more than charge.

The charge case gives a clean example of partial preservation. A black hole can preserve an exterior electromagnetic charge class even when detailed matter identity is not preserved. This is exactly what role compression means. The charge parameter is standing-bearing because it is fixed by gauge-invariant exterior response. But the equivalence class it preserves is coarse relative to arbitrary infalling microstate detail.

This prevents two opposite errors. The first error is to treat charge as merely a bookkeeping label because the interior is hidden. It is not merely bookkeeping; it changes exterior field response and thermodynamic work terms. The second error is to treat charge conservation as full information recovery. It is not that either. It is a particular exterior standing class.

Gauge discipline is essential. A gauge potential may change under redescription while the physical flux and charge class remain fixed. Therefore a potential component is not the charge role by itself. The exterior charge role is the boundary-fixed equivalence class that survives gauge-compatible redescription. This matches the general anchor/tensor/skin discipline: gauge representative skin cannot be promoted to charge tensor content.

Black-hole charge is a gauge-compatible exterior electromagnetic response. It is fixed by flux, field response, boundary conditions, and gauge-invariant comparison, not by a hidden interior possession. A Reissner-Nordstrom or Kerr-Newman

charge parameter is admissible because it changes exterior electromagnetic and metric response in a gauge-compatible way.

The charge role requires a declared gauge envelope. Gauge representatives are skin. Electric flux, boundary field response, and electromagnetic work terms are tensor or ledger content. A change of gauge does not change charge; a change in the boundary-fixed flux class does.

Theorem 9 - Black-hole charge role. A black-hole charge claim is admissible only when the charge descends through gauge-compatible boundary redescription and is fixed by electromagnetic exterior witnesses. It does not license arbitrary interior microstate standing.

Proof. A charge claim without a gauge-compatible witness is a label. A gauge presentation alone is skin. A boundary-fixed electromagnetic response or flux class has standing because it survives gauge-compatible redescription and affects exterior response. Such a role may preserve charge standing, but it certifies only the charge class. It does not reconstruct every standing-bearing detail of the infalling charged matter. Therefore charge is exterior response, not hidden interior facthood.

8.5 32. Angular momentum and black-hole spin

Angular momentum is more delicate because the word “spin” can suggest microscopic intrinsic spin. The black-hole parameter is not used here in that sense. The admissible role is rotational response of the exterior geometry and its associated boundary or asymptotic charges.

Frame dragging is an important witness because it is not merely a number in the metric. It is an exterior effect of the rotational response. Horizon angular velocity is another role: it appears in the thermodynamic work term. Asymptotic rotational charge, where defined, provides a boundary role. These are enough to make angular momentum a real exterior parameter without converting it into an interior rotating substance.

A near-extremal rotating black hole may have special thermodynamic and stability properties. Those properties enrich the role ledger. They do not remove the distinction between exterior angular-momentum response and hidden interior spin ontology.

Angular momentum is especially important because ordinary language invites an interior picture. A rotating black hole is not being described here as a material body with a hidden spinning core. The exterior angular-momentum role is fixed by response: frame dragging, asymptotic angular momentum, stationary-axisymmetric structure, horizon angular velocity, and rotational work in the first law.

The role is nevertheless physical. Frame dragging is not skin. The angular momentum parameter changes exterior geometry and affects particle motion,

wave propagation, ergoregion behavior, and thermodynamic relations. AASC does not demote angular momentum. It prevents the parameter from doing an additional unlicensed job as interior ontology.

Extremal and near-extremal cases are parameter-boundary cases. Relations among M , q_{BH} , and J can define extremality conditions in a given metric family. Those conditions are important for horizon structure and thermodynamics. They do not by themselves create an admissibility endpoint or a privileged interior fact. If an extremal model is used to claim stable storage, holographic reconstruction, or endpoint physics, the relevant certificate must still be supplied.

Black-hole “spin” is treated here as angular momentum. The paper does not claim a microscopic spin ontology for the black-hole interior. The relevant role is exterior rotational response.

The Kerr angular momentum parameter is witnessed by stationary-axisymmetric structure, asymptotic rotational charge where available, frame dragging, metric cross-terms, horizon angular velocity, and the rotational work term ΩdJ in thermodynamics. These are exterior or boundary roles. None requires the black hole to contain a rotating substance in the interior.

Theorem 10 - Black-hole angular-momentum role. A black-hole spin or angular-momentum claim is admissible when it is fixed by exterior rotational response, asymptotic or stationary-axisymmetric charge structure, frame-dragging witnesses, and continuation-covariant thermodynamic role. It is inadmissible as hidden interior rotation ontology unless a richer interior model supplies independent standing.

Proof. Angular momentum standing requires an exterior role that distinguishes rotating from nonrotating response. In a Kerr-type exterior, that role is supplied by asymptotic angular momentum, stationary/axisymmetric structure, frame dragging, and horizon angular velocity. If the same value is used as a hidden interior fact without exterior role or richer-scope certificate, it is an interior-fact upgrade. Therefore the black-hole spin parameter stands as exterior angular-momentum response.

8.6 33. Kerr-Newman exterior role package

For a stationary charged rotating exterior, define

$$\mathcal{K}_{\text{ext}} = (M_{\text{ext}}, q_{\text{BH}}, J_{\text{ext}}, H, A_{\text{wit}}, \textit{Therm}, \textit{Cert}).$$

In the standard classical Kerr-Newman representative, one often writes the rotational parameter as $a = J/M$ in geometric units, with horizon radii $r_{\pm}$ determined by the mass, charge, and angular-momentum parameters where the classical envelope admits them. The horizon angular velocity Ω_H and electric potential Φ_H enter the black-hole first-law ledger. Extremality is a parameter-boundary condition relating M , q_{BH} , and J inside that representative. None

of these formula-level roles is rejected; each is typed as exterior metric, gauge, rotational, or thermodynamic response rather than hidden interior ontology.

Component	Standard role	AASC role	Blocked overread
M	mass	source response	interior substance
q_{BH}	charge	gauge response	interior charge
J	angular momentum	rotational response	interior rotation
$r_{\pm}$	horizon radii	metric descriptors	primitive boundary
A	area	entropy ledger	microstate proof
κ	surface gravity	temperature ledger	thermal essence
Ω_H	angular velocity	rotational work	interior rotation
Φ_H	electric potential	EM work ledger	hidden charge

Theorem 11 - Kerr-Newman role normal form. The Kerr-Newman exterior package is admissible as a role package of mass, charge, angular momentum, horizon, thermodynamic, and certificate data. It classifies exterior response. It does not by itself license a truth-apt interior microstate inventory.

Proof. Each parameter and horizon quantity has an exterior or boundary role. If a proposed claim preserves those roles, it is equivalent at the exterior parameter level. If it changes them, it changes tensor or ledger content. If it adds hidden microstate labels while preserving all exterior witnesses, the addition is bookkeeping unless a richer model supplies exterior distinguishability. Therefore the Kerr-Newman package is exterior response normal form, not interior ontology.

9 Part IX - Thermodynamics and Entropy

9.1 34. Preservation of black-hole thermodynamics

The thermodynamic laws are often the strongest reason to think black holes possess microscopic structure. AASC does not resist that inference when the microscopic theory supplies the corresponding state space. It resists only the unlicensed shortcut. A thermodynamic relation among exterior quantities is a ledger. A microscopic ontology is a further model. The ledger can guide, constrain, and support the model; it is not identical to the model.

The first law also provides a useful consistency check for the parameter section. The same M , q_{BH} , and J that classify exterior response appear in the thermodynamic work terms. This strengthens the exterior role reading: black-hole parameters are not arbitrary labels. They participate in metric, field, and thermodynamic ledgers. Their strength as exterior data is precisely why they should not be overread as interior microstate inventory.

The first-law ledger also clarifies the relation between mechanics and thermodynamics. In classical black-hole mechanics the area, surface gravity, angular

velocity, and electromagnetic potential obey laws formally analogous to thermodynamic laws. Semiclassical Hawking radiation gives temperature physical force in the usual thermodynamic sense. AASC does not contest that development. It asks which role each quantity occupies in the black-hole reconstruction problem.

The term $T dS$ is especially prone to overpromotion. It can be a legitimate thermodynamic term without serving as a microstate reconstruction map. An area law may be evidence that a deeper microphysical model counts states. A successful string-theoretic or holographic state count may explain an entropy value in a declared scope. But the entropy scalar itself does not identify which infalling reference class is later re-fixed by exterior witnesses.

Likewise, the work terms ΩdJ and Φdq_{BH} preserve important exterior parameter standing. They show how angular momentum and charge participate in the thermodynamic ledger. They do not imply that the exterior has recovered every hidden interior degree of freedom compatible with those parameters.

This role discipline allows black-hole thermodynamics to remain strong. It is not weakened by refusing entropy-to-ontology promotion. On the contrary, it becomes clearer: thermodynamics controls exterior ledger relations; information reconstruction requires certificate structure; microstate counting requires a declared microphysical model.

Black-hole thermodynamics is preserved. The AASC claim is not that entropy, temperature, area, or the first law are invalid. The claim is that they require ledgers. They do not become primitive ontology merely because they are powerful and successful.

Following the standard black-hole mechanics and thermodynamic interface [9-11], in stationary form the first-law expression is

$$dM = T dS + \Omega dJ + \Phi dq_{\text{BH}}.$$

This equation is role-rich. The term dM is the exterior mass/source-response ledger. The term $T dS$ is the temperature-area or entropy ledger. The term ΩdJ is the rotational work and angular-momentum ledger. The term Φdq_{BH} is the electromagnetic work and charge ledger. The equation is a relation among exteriorly licensed thermodynamic roles.

The manuscript does not derive the first law, Hawking temperature, or Bekenstein-Hawking entropy. It classifies their admissibility status. They are valid thermodynamic ledgers within their declared physical scope. They do not by themselves prove that every interior microstate is a truth-apt exterior-standing fact.

9.2 35. Entropy types and ledgers

The entropy distinctions are not optional. Bekenstein-Hawking entropy is geometric and thermodynamic. Entanglement entropy is algebraic and partition-relative.

Generalized entropy mixes area and matter terms. Statistical entropy counts states relative to an equivalence relation. Coarse-grained entropy tracks operationally inaccessible distinctions. These objects often interact, but they are not interchangeable.

A black-hole information argument that moves from one entropy type to another must declare the bridge. For example, using generalized entropy to derive a Page curve involves a reconstruction and extremization setup. Using string microstate counting to match area entropy involves a microphysical counting problem. Using entanglement entropy to infer recoverability requires an algebra and map. AASC does not object to any bridge that is declared. It objects to bridges hidden inside the word “information.”

The distinction among entropy types also matters for islands. Generalized entropy combines an area contribution with quantum entropy of fields in a specified region. In island calculations the extremization of generalized entropy can identify a reconstruction region. AASC reads this as a candidate certificate mechanism only when the region and algebra are tied to exterior reconstruction. Generalized entropy is not automatically an exterior witness; it is part of a reconstruction ledger.

Entanglement entropy also requires caution. It depends on a partition, algebra, cutoff or renormalization structure, and state. In quantum gravity those ingredients can be subtle. A claim that entanglement entropy preserves information must say which algebra carries the standing-bearing distinctions and how that algebra is available to the exterior scope.

State-counting entropy has a different status. If a string theory, fuzzball model, or holographic theory counts microstates corresponding to a black hole, that is a powerful microphysical result. AASC then asks whether the counted distinctions affect the declared exterior reconstruction problem. A state count can explain entropy while still not providing exterior distinguishability for every state. Conversely, a holographic reconstruction map might preserve exterior standing without requiring a naive local interior microstate picture.

Thus entropy supports several legitimate roles. It can be thermodynamic, generalized, entanglement-based, microstate-counting, or coarse-grained. The manuscript rejects only the move from any one of these roles to primitive interior facthood without the missing ledgers.

Entropy can be strong evidence for microphysical structure without becoming primitive ontology. Matching Bekenstein-Hawking entropy by state counting, deriving a generalized-entropy extremization rule, or computing an entanglement entropy can all support a physical model. The AASC restriction is narrower: the entropy result must still say which carrier, algebra, equivalence relation, and reconstruction map make the counted or extremized distinctions standing-bearing for the declared exterior problem. Entropy may therefore motivate and constrain microphysics without automatically licensing globally referenceable interior facthood.

Black-hole entropy appears in several registers.

Bekenstein-Hawking entropy is an area/thermodynamic ledger. It relates horizon area to thermodynamic entropy in the appropriate semiclassical regime.

Generalized entropy combines geometric area terms with matter or exterior entropy terms in semiclassical and reconstruction contexts. It is indispensable in island and quantum-extremal-surface programs, but it belongs to a declared reconstruction or semiclassical ledger.

Entanglement entropy is algebra, subsystem, or partition dependent. It requires a state, a subsystem or algebra, and a trace or restriction operation.

State-counting entropy belongs to a microphysical model. It can be a powerful explanation if the microstates, equivalence relation, counting measure, and exterior relevance are declared.

Coarse-grained exterior entropy belongs to operational or thermodynamic exterior descriptions. It may classify macroscopic ignorance or accessible records without asserting microscopic interior facthood.

Theorem 12 - Entropy-ledger theorem. A black-hole entropy claim is admissible only relative to a declared state space, algebra, coarse-graining, boundary area relation, thermodynamic scope, or microphysical counting ledger. Entropy cannot by itself supply interior facthood or exterior standing transport.

Proof. An entropy assignment requires a target set or state, an equivalence relation or coarse-graining, a measure or algebra, and a comparison ledger. In the black-hole setting, these are supplied by thermodynamics, semiclassical gravity, algebraic reconstruction, or a microphysical model. If the ledger is absent, the entropy scalar is not a standing-bearing interior ontology. If the ledger is present, entropy has standing only in that ledger. Therefore entropy is preserved as ledger content but cannot alone transport exterior reference.

9.3 36. Entropy-to-ontology promotion

Entropy often motivates microstate ontology. AASC does not prohibit microstate models. It prohibits an unlicensed inference from entropy to truth-apt interior facthood.

Theorem 13 - Entropy-to-ontology promotion boundary. If black-hole entropy is used to infer truth-apt interior microstates, the argument must supply a microstate carrier, an equivalence relation, an encoding or counting map, an exterior or boundary witness relation, redescription invariance, and a reconstruction or standing-transport connection. Absent those fields, entropy is a thermodynamic ledger, not an interior ontology proof.

Proof. A microstate claim identifies objects, counts or distinguishes them, and treats those distinctions as physically meaningful. Without a carrier there is no target. Without an equivalence relation there is no stable identity. Without a

counting or encoding map entropy has no microstate content. Without exterior or boundary witness relation the distinctions do not affect the declared reconstruction scope. Without redescription invariance the microstates are presentation artifacts. Without reconstruction or standing transport they do not preserve threatened exterior reference. Therefore entropy can support a microphysical theory only when these ledgers are supplied.

10 Part X - Information Loss and Exterior Reconstruction

10.1 37. Precise information-loss target

The paper avoids the broad phrase “information loss is inadmissible” because information has many meanings. The precise target is this:

Uncertified loss of standing-bearing reference in a declared exterior reconstruction scope is inadmissible as continuing same-scope preservation.

Information as Shannon quantity, von Neumann entropy, Hilbert-space norm, Page-curve behavior, holographic encoding, remnant capacity, or algorithmic recoverability is not enough by itself. Each becomes relevant only through carriers, encodings, exterior witnesses, and reconstruction maps.

Term	Not enough for standing	Sufficient only if
Shannon information	encoding-dependent quantity	carrier and witness ledger fixed
von Neumann entropy	subsystem or algebra dependent	reconstruction role declared
Hilbert norm	global predictive structure	exterior standing map induced
Page curve	entropy or correlation evidence	supports certificate for declared exterior references
Holographic encoding	dual representation	anchor map to exterior witness algebra
Remnant capacity	storage space	exterior remnant map exists
Global unitarity	total evolution	exterior recoverability certificate induced

10.2 38. Pure-to-mixed evaporation and Page behavior

A pure-to-mixed description in semiclassical evaporation is a statement about the predictive or density-matrix ledger of the exterior theory. It can be physically

and conceptually serious. AASC does not dissolve it by word choice. It refines the target. If the exterior density matrix loses access to distinctions that the declared reconstruction problem requires, then a standing loss has occurred in that scope. If a later theory restores those distinctions through correlations, boundary reconstruction, or an enlarged algebra, then the loss has been avoided or reclassified.

The Page curve is important because it diagnoses whether radiation entropy behaves as expected under unitary evaporation. But entropy behavior and reference re-fixation are different. If a Page curve is accompanied by an explicit reconstruction map, it becomes certificate evidence. If not, it remains a strong constraint on candidate theories rather than a completed standing-transport proof.

This distinction also handles approximate recovery. A physical theory may recover reference only up to a code subspace, coarse-graining, error tolerance, late-time limit, or operational class. AASC can admit such recovery if the scope declares the tolerance and standing quotient. What it cannot admit is a shift between exact and approximate targets without saying so.

In the semiclassical formulation, one may describe a pure incoming state and apparently thermal outgoing radiation, producing the familiar pure-to-mixed tension. AASC does not deny that this is the standard pressure point. It asks which standing-bearing reference classes are lost, which exterior witnesses remain, and whether a reconstruction certificate is supplied.

A Page curve, if derived in a model, is important evidence about entropy and correlation behavior. It is not automatically a standing certificate. It becomes AASC-relevant when the recovered correlations support a redescription-stable map from threatened reference classes to exterior witnesses. Otherwise it is an entropy ledger without a declared reference re-fixation map.

Thus the AASC information-loss theorem neither denies nor trivializes Page-curve work. It classifies what the Page curve must become for the exterior reconstruction problem: not merely an entropy plot, but evidence for a recoverability certificate.

10.3 39. Global retention and exterior recovery

The recoverability certificate Rec_G is deliberately stronger than a statement of global existence. It must land in the old exterior witness algebra. This landing requirement is what separates AASC from slogans about information being somewhere. A global code may contain the data. A boundary theory may contain the data. A remnant may contain the data. A baby universe may contain the data. None of these facts by itself says that the old exterior scope can re-fix the reference class.

The certificate also prevents representative privilege. A chosen reconstruction dictionary cannot be introduced after seeing the desired answer unless it is part

of the declared scope. A final-state map cannot select the missing recovery channel unless it is an anchor. A Page-curve inference cannot identify which reference class is recovered unless the correlation map is declared. The theorem demands not only existence of structure, but role-fixed structure.

Let G be a larger-domain description: a global wavefunction, boundary dual state, exact quantum-gravity state, island-extended reconstruction object, remnant Hilbert space, or baby-universe state. Let $G|_{S_{\text{BH}}}$ denote its restriction or projection to the declared exterior black-hole scope. An exterior recoverability certificate is a map

$$Rec_G : G|_{S_{\text{BH}}} \rightarrow A_{\text{wit}}$$

satisfying anchor re-fixation, redescription stability, continuation covariance, exterior availability, no hidden selector, no post-hoc repair, and preservation of Q_{ext} .

Theorem 14 - Global retention separation. A larger-domain retention claim preserves exterior standing in S_{BH} if and only if it induces an exterior recoverability certificate Rec_G satisfying the listed conditions.

Proof. If Rec_G exists, then threatened exterior reference classes are re-fixed by admitted exterior witnesses or reconstruction maps. Redescription stability and continuation covariance ensure the map preserves standing rather than selecting representatives. Exterior availability ensures the map belongs to the declared exterior reconstruction problem. No-selector and no-repair clauses prevent acquisition of standing outside the old scope. Thus exterior standing is preserved.

Conversely, if a larger-domain claim preserves exterior standing, then for every threatened exterior class it must determine an exterior witness image preserving the standing quotient. The collection of these assignments is precisely a recoverability certificate. If no such map exists, the larger-domain data may retain state, correlation, or norm, but it does not preserve exterior standing. Treating it as decisive would be hidden global indexing, scope change, or bookkeeping.

10.4 40. Applications of the global-retention theorem

The theorem also explains why different programs can appear to disagree while addressing different scopes. A global quantum-gravity model may insist that the total state is pure. A semiclassical exterior calculation may produce a thermal density matrix. A holographic calculation may reconstruct an interior operator from boundary data. These statements are not automatically contradictory because they refer to different carriers and algebras. The conflict arises only when one is used to satisfy the standing burden of another without a map.

For example, a boundary theory in AdS/CFT can be unitary and complete relative to its own Hilbert space. To serve as an exterior black-hole certificate, it

must identify the relevant black-hole exterior scope and show how threatened exterior references map to boundary observables or code-subspace reconstructions. In many holographic contexts this is precisely the kind of structure being developed. AASC treats it as the right kind of answer, not as a failure.

By contrast, a remnant claim that only states “the information is stored in the remnant” supplies no Rec_G . A remnant map may be possible, but it must be shown. Similarly, a baby-universe claim may preserve global information in a multiverse-like domain while failing to give the old exterior any standing image. The theorem distinguishes retention somewhere from recovery in the declared scope.

A **global wavefunction** may preserve norm or purity. It preserves exterior standing only through a map into the exterior witness algebra.

AdS/CFT or holographic unitarity is a strong certificate candidate because it supplies boundary dynamics and reconstruction technology. It becomes same-scope exterior recovery when it maps threatened exterior reference classes into the declared boundary or exterior algebra.

An **island construction** may extend the reconstruction domain. It succeeds for AASC if the island prescription factors through the declared exterior algebra and preserves Q_{ext} . If it silently changes the scope while presenting the result as old-scope exterior recovery, it is scope transport.

A **remnant Hilbert space** has capacity. Capacity is not standing. A remnant preserves exterior standing only if an exterior remnant map identifies how threatened classes are re-fixed.

A **baby universe** may retain state in a larger domain. It preserves old exterior standing only if it induces an exterior image.

A **final-state projection** can be treated as a declared physical boundary condition only if it is present as an anchor in the model. If it is introduced to repair undefined transport after the fact, it is a selector.

11 Part XI - Major Program Interfaces

The programs in this part are not rejected. They are evaluated by the certificate they supply relative to the declared exterior reconstruction scope. A program may satisfy the theorem by providing leakage, a boundary algebra, a reconstruction map, a non-sink quotient, or a declared richer scope. The AASC question is not whether the program is mathematically interesting; it is whether it re-fixes the threatened exterior reference class in the old or declared enlarged scope.

11.1 41. Semiclassical evaporation

Contribution. Semiclassical evaporation supplies the most direct leakage candidate [11,12]. It gives an exterior radiation channel, energy flux, entropy

behavior, and late detector records in a metric quantum-field-theoretic envelope. AASC does not derive the Hawking temperature, greybody factors, or evaporation time; it classifies what role radiation must play if invoked against the standing-sink obstruction.

AASC success condition. Radiation must carry, or be linked by a declared map to, the threatened exterior reference classes at issue. A flux can lower mass and support thermodynamics without reconstructing the detailed reference class of an infalling probe. Standing recovery requires a radiation-to-reference relation.

Failure mode. A purely thermal ledger with no reconstruction relation may be physically important but does not preserve standing-bearing exterior reference. An endpoint that restores reference without declaring a mechanism is repair.

Certificate form. The certificate is a radiation-to-reference map, which may be approximate, asymptotic, algebraic, or model-specific. The theorem requires declaration of the map, not a single favored dynamics.

11.2 42. Holography

Contribution. Holography is one of the strongest possible responses to the horizon theorem [15-17]. A boundary theory, code subspace, entanglement wedge, or operator reconstruction map can supply exactly the sort of anchor transport AASC requires.

AASC success condition. The holographic map must preserve the exterior standing quotient Q_{ext} , the admitted witness algebra A_{wit} , and the relevant continuation relation. If the map is code-subspace relative, state-dependent, or restricted to a low-energy algebra, the standing claim is restricted accordingly.

Failure mode. The failure mode is dictionary laundering: treating the phrase “encoded on the boundary” as sufficient without showing how the boundary object re-fixes the threatened exterior classes. If the asymptotic boundary or algebra changes, the scope has moved.

Certificate form. The certificate is a boundary-algebra or code-subspace reconstruction map into the exterior witness algebra, together with the redescription discipline that preserves the relevant quotient classes.

11.3 43. Islands and replica wormholes

Contribution. Island and replica-wormhole methods provide generalized-entropy and reconstruction-domain machinery [22-26]. They are serious certificate candidates because they can identify a region reconstructable from the radiation algebra and reproduce Page-curve behavior.

AASC success condition. The island prescription succeeds for the present theorem only when the resulting reconstruction factors through the declared

exterior or radiation algebra. Then the island is not a magic interior fact imported into the exterior; it is part of a certified reconstruction map.

Failure mode. The failure mode is scope transport presented as old-scope recovery. If an interior region is simply added to the exterior description without the algebraic or path-integral rule that makes it exteriorly reconstructable, the old exterior witness algebra has not recovered the standing class.

Certificate form. The certificate is an algebraic, path-integral, or reconstruction map showing how the radiation/exterior algebra re-fixes the threatened class while preserving the declared quotient and continuation profiles.

11.4 44. Complementarity and firewalls

Contribution. Complementarity contributes a way to relate exterior and infalling descriptions without requiring both to be simultaneously available to one observer [14,20]. Firewall models contribute a possible horizon tensor-load response to entanglement and smoothness constraints.

AASC success condition. Complementarity succeeds only if it supplies redescription-stable correspondence preserving the standing-bearing content being compared. Firewall models succeed as physical models only when the firewall is declared as tensor content of the horizon regime rather than inferred from inconsistent bookkeeping.

Failure mode. Complementarity fails as same-scope exterior reconstruction if it only changes observer scope. Firewall pressure becomes overread when incompatible global bookkeeping demands are imposed across a fixation boundary and then treated as proof of a new object without declaring the corresponding physical load.

Certificate form. Complementarity requires an exterior/infalling correspondence map. A firewall requires a declared horizon-tensor model with its source, witness, and continuation status fixed.

11.5 45. Remnants, fuzzballs, final states, and baby universes

Contribution. Remnants offer late storage; fuzzballs and microstate geometries may replace the classical interior; final-state projection introduces an endpoint boundary condition; baby-universe routes move retention into a larger domain [18,19]. These are distinct physical roles, not one generic rescue.

AASC success condition. A remnant must expose an exterior remnant map. A fuzzball or microstate-geometry model succeeds when its microstate distinctions have exterior, boundary, thermodynamic, or reconstruction standing in the declared target. Final-state projection succeeds only when the final condition is a declared physical anchor from the start. A baby-universe route

succeeds for the old exterior only if it induces a map back to the old exterior quotient.

Failure mode. Remnant capacity without exterior distinguishability is a hidden compression sink. Fuzzball labels with no witness effect are hidden labels. A final-state condition added after loss is selector repair. A baby universe without an induced old-exterior image is scope transport.

Certificate form. The corresponding certificates are, respectively, a remnant-to-witness map, exterior microstate witnesses or reconstruction maps, an independently licensed final-state boundary condition, and a branch-to-exterior standing map.

12 Part XII - Detailed Interfaces and Objections

12.1 46. Formation, collapse, and the exterior history

A black-hole reconstruction problem normally arises from a history, not from an eternal diagram alone. Matter begins in an exteriorly describable state, gravitational collapse creates a trapped or horizon-like region, exterior fields settle toward a parameterized geometry, radiation may be emitted, and late records are compared with early preparation. AASC treats this sequence as an exterior standing-history problem. The question is not whether every stage has an exact classical description. The question is which reference classes remain re-fixable through the declared exterior witnesses.

In a collapse scenario, an infalling shell or object has exterior standing before crossing a horizon-like boundary. Some standing-bearing roles may remain exterior: total mass, charge, angular momentum, and perturbative response. Other distinctions may not. If the model later claims that all distinctions are preserved, it must say how the exterior obtains them. If it claims only coarse exterior parameters survive, then the theorem is not offended; the target has been narrowed. The information paradox becomes sharp only when the theory also demands preservation of distinctions not available in the old exterior witness algebra.

This is why the Horizon Isolation Lemma is stated for threatened classes rather than for everything that falls in. AASC does not force a theory to preserve every arbitrary representative detail. It asks that any standing-bearing reference class included in the declared reconstruction burden have a future exterior image. The burden is declared by the problem itself: formation histories, unitarity claims, microstate counts, dual reconstructions, and information-recovery statements all specify which classes they intend to preserve.

Eternal black-hole solutions occupy a different place. They may be exact classical metrics with no formation history in the model. They can serve as thermodynamic backgrounds, scattering geometries, or limiting representatives. But an eternal horizon cannot be used as a silent proof that formation-history reference classes

have no reconstruction burden. Conversely, the horizon obstruction does not invalidate the eternal solution. It only says that when the metric is used as a permanent terminal boundary for standing-bearing exterior reconstruction, certificate discipline becomes necessary.

12.2 47. Asymptotic flatness, AdS boundaries, and scope differences

The exterior witness algebra depends on the boundary regime. In asymptotically flat settings, exterior witnesses may include null-infinity data, Bondi flux, ADM quantities, scattering states, gravitational memory, and finite detector records. In AdS-like settings, boundary correlators, boundary Hamiltonian evolution, and holographic reconstruction can play a more explicit role. These are not interchangeable scopes.

A holographic AdS black hole may have a boundary description that is far richer than the exterior algebra in an asymptotically flat semiclassical collapse model. If a boundary theory supplies complete unitary dynamics and operator reconstruction, then it may satisfy the certificate burden. But that success belongs to the declared boundary scope. Transporting it into another scope requires a map between the witness algebras.

This is one reason the theorem is not stated as a universal claim that all horizons evaporate. It is stated as a fixed-domain reconstruction theorem. In a scope where the boundary theory supplies the required certificate, the horizon is not uncertified. In a scope where the exterior algebra lacks such a certificate, global confidence in a larger theory does not by itself define old-scope transport.

The same issue arises for finite boxes, cosmological horizons, and observer-relative horizons. A Rindler horizon, de Sitter horizon, trapping horizon, and astrophysical black-hole horizon do not automatically share the same witness algebra or standing burden. AASC classifies them only after the scope is declared. The black-hole theorem applies when the horizon functions as a permanent obstruction to standing-bearing exterior reconstruction in a fixed scope.

12.3 48. Exact, effective, and approximate reconstruction

Physical reconstruction is often approximate or effective. A late radiation state may reconstruct early data only within a code subspace; a detector may recover only coarse-grained parameters; a semiclassical approximation may ignore exponentially small correlations; an operational protocol may require asymptotic time. AASC allows such structures when the scope declares them.

The standing quotient can be adjusted to the intended exactness. If the target is exact microstate recovery, then the witness algebra must distinguish exact classes. If the target is recovery up to a code subspace, the quotient should identify states outside the code. If the target is thermodynamic recovery, coarse-grained

entropy and response ledgers may suffice. If the target is exterior parameter preservation, M , q_{BH} , and J may be enough.

The inadmissible move is not approximation. It is target drift. A calculation may recover coarse thermodynamic data and then be reported as full microstate recovery. A Page-curve result may support entropy-unitarity behavior and then be reported as reference reconstruction. A boundary code may reconstruct a low-energy operator algebra and then be reported as complete interior facthood. AASC requires that the reported target match the declared quotient.

This exactness discipline prevents both underclaim and overclaim. It allows partial success to count as partial success. It also prevents a partial success from silently filling the role of a stronger theorem. In black-hole information physics, this is a practical virtue: many results are extremely valuable because they solve a well-typed subproblem. AASC protects those subproblems from being overburdened.

12.4 49. Exterior quotient compression and hair-role typing

Exterior quotient compression is often interpreted as a loss of information: many infalling configurations can share the same exterior response package. AASC treats this as exterior role compression. The declared exterior quotient preserves exactly the standing-bearing distinctions it re-fixes: in the classical settled example, mass, charge, angular momentum, and any additionally declared exterior witnesses or charges. It does not preserve every detailed standing-bearing distinction unless a reconstruction mechanism supplies them elsewhere.

This compression can be physically real without being paradoxical. Many physical processes lose operational distinctions in a coarse exterior description. The paradox arises when another part of the theory demands that finer distinctions remain recoverable in the same or a related exterior scope. In that case, the exterior quotient is insufficient for the stronger reconstruction burden, but it is not false. It is simply too coarse for the finer target.

Alleged hair is handled by role typing. If the hair changes exterior observables or a boundary algebra in a standing-bearing way, it is tensor, witness, parameter, or certificate content and subtracts only what it re-fixes from the unresolved residue. If it is invisible to all exterior witnesses but still asserted as standing-bearing, it is hidden load. If it resides in a larger domain, the map back to old exterior standing is required. If it changes no standing-bearing fact, it is skin or bookkeeping. Thus hair-language supplies no independent route around exterior quotient subtraction.

Exterior quotient compression also clarifies why mass, charge, and angular momentum are not hidden interior facts. They are response classes that survive exterior compression. Their survival is exteriorly meaningful, but their survival

does not imply survival of all other distinctions. The normal form respects that asymmetry.

12.5 50. Inner horizons, Cauchy horizons, and determinacy

Charged and rotating black-hole solutions can contain inner horizons or Cauchy horizons in their classical analytic extensions. These structures raise questions about determinacy, stability, and mass inflation. AASC classifies them under continuation and high-curvature discipline rather than treating them as ordinary exterior event horizons.

A Cauchy horizon concerns whether data on an earlier surface determine a later region in the metric model. If instability or mass inflation destroys the continuation, the result is a failure or high-curvature diagnostic. If a model claims stable continuation through the Cauchy horizon, it must supply the relevant continuation and source ledgers. If it uses the region to store or transmit standing-bearing content, it must supply the exterior certificate or declare a richer scope.

This classification avoids two errors. It does not dismiss inner-horizon physics as irrelevant. Inner horizons are important in classical and perturbative black-hole theory. But it also does not allow analytic extension to become standing transport. A mathematical extension can display regions beyond a Cauchy horizon. Exterior standing preservation requires more than display; it requires a witness or certificate relation.

The same caution applies to chronology-violating regions in extended Kerr-like diagrams. A representation containing such regions is not automatically a physical claim about accessible interior ontology. It is a metric extension whose physical standing depends on the declared envelope and continuation discipline.

12.6 51. The equivalence principle and local smoothness at the horizon

AASC does not deny the local smoothness expectation at a large classical horizon. A freely falling observer may experience no local drama at a sufficiently large horizon in semiclassical effective field theory. This is a local metric and field-theoretic statement inside a declared approximation. It does not by itself solve exterior reconstruction.

The firewall debate arises partly because local smoothness, exterior unitarity, entanglement monogamy, and interior mode assignments are placed in tension. AASC treats the conflict as a scope and role problem. Which algebra carries the interior mode? Which exterior witnesses preserve the reference? Which redescription correspondence relates infalling and exterior accounts? Which assumptions are exact and which are effective?

If the declared model supplies a horizon tensor load, then a firewall is a physical feature in that model. If it supplies complementarity correspondence, the conflict may be dissolved by redescription. If it supplies holographic reconstruction, the interior mode may be encoded differently. If it supplies none of these and still demands incompatible global bookkeeping, the contradiction is generated by overpromoted assumptions.

The equivalence principle therefore remains respected as a local metric principle in its domain. It is not forced to carry the global exterior reconstruction burden by itself.

12.7 52. Algebraic formulation of the certificate condition

The certificate condition can be stated algebraically. Let $\mathcal{A}_{\text{ext}}$ be the declared exterior algebra, $\mathcal{A}_{\partial H}$ the boundary-trace algebra where present, and $\mathcal{C}$ the certificate structure. A threatened standing class $[r] \in Q_{\text{ext}}$ is preserved if there exists an admissible assignment

$$C_r : [r] \mapsto a_r \in \text{Alg}(\mathcal{A}_{\text{ext}}, \mathcal{A}_{\partial H}, \mathcal{C})$$

such that a_r re-fixes the class, is stable under admissible redescription, and respects continuation. The details of the algebra depend on the physical model. The certificate condition abstracts the required role.

In a semiclassical radiation model, a_r may be a late radiation correlation. In holography, it may be a boundary operator or code-subspace reconstruction. In an island calculation, it may be a radiation-algebra reconstruction involving a generalized entropy prescription. In a remnant model, it may be an exterior remnant observable. In a final-state model, it may be a boundary-conditioned map.

This abstraction is intentionally permissive. It does not require one preferred black-hole solution. It requires that the solution be typed by its reconstruction role. If no such a_r exists, then the class is not preserved in the declared exterior algebra.

12.8 53. Why authority, calculability, and elegance do not generate standing

A result can be technically beautiful and still fail to supply the required certificate. A duality can be powerful while not addressing a particular exterior scope. A numerical simulation can be accurate within a model while not fixing a reference class. A Page curve can be a decisive entropy result while not identifying the recovered reference. A remnant capacity estimate can be large while not exposing exterior retrieval.

AASC therefore separates performance from standing. Performance matters when it induces a standing-preserving map. Calculability matters when the calculation is about the declared quotient. Elegance matters rhetorically but has no standing force by itself. Authority, consensus, or expectation cannot turn an undefined old-scope transport assignment into a defined one.

This discipline is not anti-physics. It is how physical claims remain clear. When a program supplies a certificate, AASC recognizes it. When a program changes scope, AASC records the new scope. When a program gives evidence for a certificate but not the certificate itself, AASC records the evidence without overreporting it.

12.9 54. Boundary of the present theorem

The present theorem is a structural reconstruction theorem. It does not predict astrophysical waveforms, compute greybody factors, derive entropy coefficients, select a quantum-gravity completion, or determine whether a given evaporating black hole leaves a remnant. Those are physical and mathematical tasks in their own right.

The theorem does constrain how such results can be used. A waveform can witness exterior parameter response. A greybody factor can contribute to a radiation ledger. An entropy coefficient can contribute to thermodynamic or state-counting ledgers. A quantum-gravity model can supply a certificate. A remnant model can expose an exterior map. None of these results can silently perform the role of another.

The boundary is therefore sharp. AASC does not do the dynamics. It fixes the admissibility grammar of the claims those dynamics may support. A completed physical model remains necessary. But once a model is presented as solving a black-hole reconstruction problem, the certificate and scope requirements apply.

12.10 55. Case study: collapse with parameter preservation but no microstate certificate

Consider a collapse model in which an exterior distribution of matter settles to a stationary exterior with mass M_{ext} , charge q_{BH} , and angular momentum J_{ext} . The exterior parameter package is preserved. The exterior geometry, electromagnetic field, and rotational response may be accurately described by a Kerr-Newman exterior after ringdown. This is a success for exterior parameter standing.

Now add a stronger claim: every distinguishable preparation of the collapsing matter remains recoverable by a late exterior observer. That stronger claim is not supplied by the parameter package. Many possible preparations can share the same $(M_{\text{ext}}, q_{\text{BH}}, J_{\text{ext}})$. The exterior quotient compresses them. To preserve the stronger reference classes, the model must supply radiation correlations, boundary data, microstate geometry, or another certificate.

This case shows how AASC avoids both exaggeration and dismissal. It does not say that the exterior parameters are meaningless. They are standing-bearing exterior roles. It does not say that microstate recovery is impossible. It says that microstate recovery is a further claim requiring a further map. A theory that supplies such a map is doing real additional work beyond classical exterior compression.

The same case also explains why thermodynamics alone is not enough. The final area and entropy may encode a coarse thermodynamic count. They do not identify which preparation has which late exterior image. A state-counting model can add that bridge; a thermodynamic equation by itself does not.

12.11 56. Case study: holographic recovery as certified totalization

A holographic model can be read as a positive example of the theorem’s allowed route. Suppose the boundary theory supplies a Hilbert space, a boundary algebra, a code subspace, and a reconstruction map for the black-hole exterior problem. Suppose further that the map identifies the threatened exterior reference classes, preserves their equivalence under redescription, and remains compatible with the declared continuation relation. Then the horizon standing-transport operator is not being illicitly totalized. The boundary map supplies a certified domain extension or a certificate inside the declared scope.

In such a case, AASC does not object that the classical interior was inaccessible. The point of the certificate is precisely to provide a lawful exterior or boundary image. Holography succeeds, relative to the theorem, by preventing the old horizon from functioning as an uncertified standing sink.

The same example also illustrates scope sensitivity. The boundary theory may be complete in an AdS setting. A physically different asymptotically flat collapse problem does not automatically inherit the same certificate. A map between the scopes is required. If supplied, it can be part of the theorem object. If not, importing the AdS certificate into a different exterior problem is scope transport.

Thus the theorem is not anti-holographic. It is a grammar for saying exactly what holographic recovery accomplishes: it supplies, or attempts to supply, an exterior-standing certificate.

12.12 57. Case study: remnant storage without exterior map

A remnant scenario can be stated in a weak or strong form. The weak form says a late object remains after evaporation. The strong form says that this object stores the standing-bearing content threatened by the horizon. The weak form is a claim about endpoint dynamics. The strong form is a claim about reconstruction.

For AASC, storage is not the same as standing. A remnant may have a large internal Hilbert space, a complicated microstructure, or a long lifetime. None of those facts preserves old exterior standing unless there is an exterior remnant map. The map must identify how late exterior witnesses distinguish or recover the old reference classes.

If the remnant has no such exterior map, then it is a compression sink relative to the old exterior reconstruction problem. It may be a meaningful object in a larger theory, but it is not a same-scope rescue. If the map is supplied, then the remnant becomes a certificate-bearing route and is not excluded by the theorem.

This case is structurally parallel to charge preservation. Charge is a successful exterior map for one role. A remnant capacity without a retrieval map is not. The difference is not the size of the carrier; it is whether the standing relation is exposed to the declared witnesses.

12.13 58. Case study: island reconstruction and target declaration

Island reconstruction is subtle because it changes what is counted as reconstructable from radiation. A naive reading might say that the island is inside the black hole and therefore cannot be part of the exterior. A more careful reading says that the radiation algebra, in the gravitational path-integral calculation, reconstructs a region that includes the island. AASC can accommodate the second reading.

The target declaration is decisive. If the declared theorem object includes the island prescription as part of the reconstruction map from radiation to bulk operators, then the island can be a certificate mechanism. The exterior standing class is preserved not because the classical interior became directly observable, but because the declared reconstruction algebra maps the class into the radiation or boundary description.

If, however, the island is introduced merely as a region whose interior facts are granted to the exterior without an algebraic or path-integral certificate, then it is scope transport. The classical exterior problem has been enlarged without declaring the map. AASC does not reject islands; it requires the reconstruction role to be stated.

This case shows why the paper separates metric interior from reconstruction interior. The metric location of an island is not the same as its standing role. A region may be geometrically interior but reconstructively exterior relative to a declared algebra. That is admissible when the algebraic certificate is supplied.

12.14 59. Case study: final-state projection and claim order

Final-state projection illustrates the importance of claim order. A final-state condition declared as part of the theory from the beginning is an anchor. It may define a lawful boundary condition for the model. Its consistency, unitarity, and physical plausibility are then evaluated as part of that model.

The same condition introduced only after a standing-transport failure has appeared plays a different role. It makes an old undefined assignment defined after the fact. In AASC terms, that is repair unless the condition was already part of the declared scope or is introduced as a scope change.

This distinction is not semantic. A boundary condition changes the theorem object. If it is present from the start, the horizon may be certified by it. If it is added later while pretending the old object has not changed, it hides the source of reconstruction. The final-state proposal therefore becomes admissible only with explicit anchor status.

This case generalizes to any endpoint rule. A rule at a singularity, remnant, inner horizon, or Planck-scale region must be declared as physics. It cannot appear as an invisible repair site.

12.15 60. Certificate checklist for black-hole reconstruction claims

A reconstruction claim is complete only when it answers the following questions.

1. What is the exterior scope? The answer fixes S_{BH} and prevents scope drift.
2. Which reference classes are being preserved? The answer identifies the relevant elements of R_{ext} .
3. What is the standing quotient? The answer fixes which distinctions count.
4. What is the witness algebra? The answer identifies A_{wit} .
5. Where does the horizon act? The answer decomposes X_H into exterior, boundary, and isolated carriers.
6. What is the certificate? The answer gives leakage, radiation, boundary trace, dual map, island prescription, complementarity correspondence, remnant map, or quotient.
7. How is redescription stability preserved? The answer prevents representative selection.
8. How is continuation covariance preserved? The answer prevents history reset.
9. What are the parameter and thermodynamic ledgers? The answer prevents mass, charge, spin, or entropy from doing untyped work.
10. What is the failure mode if the certificate is absent? The answer classifies exhaustion, scope change, selector import, repair, or fail-closed status.

This checklist is not an external methodological preference. It is the operational

version of the normal-form theorem. A black-hole program can be very different in its mathematics and still pass the checklist. Semiclassical radiation, holography, islands, fuzzballs, remnants, and final-state models can all supply answers. What fails is the absence of answers while preservation is still claimed.

12.16 61. Relation to empirical black-hole astronomy

The theorem is structural, but it is compatible with empirical black-hole astronomy. Gravitational-wave ringdowns, black-hole shadows, accretion signatures, orbital dynamics, and electromagnetic observations provide exterior witnesses for mass, angular momentum, compactness, horizon-like behavior, and deviations from classical metrics. These are precisely the kinds of data that populate A_{wit} for astrophysical contexts.

Such observations do not normally attempt full microstate reconstruction. They test exterior parameter packages, metric descriptors, and environmental models. AASC therefore classifies them as exterior observable and parameter-response witnesses. If an observation constrains echoes, near-horizon structure, or deviations from Kerr, it may also constrain possible horizon certificate or microstructure models.

The theorem does not claim that astrophysical observations presently decide the information paradox. It says that empirical data enter through exterior witness roles. A black-hole model that claims empirical support must say which role the observation supports: mass, spin, charge, horizon compactness, thermodynamic behavior, radiation, or reconstruction.

This empirical compatibility matters because it prevents the manuscript from becoming purely abstract. AASC does not replace observation with classification. It clarifies how observations bear standing: they fix exterior roles, not hidden ontology by default.

12.17 62. Relation to quantum field theory in curved spacetime

Quantum field theory in curved spacetime provides the semiclassical setting for Hawking radiation. It treats fields on a fixed or slowly varying background and derives particle creation, stress-energy expectation values, and thermodynamic behavior. AASC does not replace that calculation. It classifies what the calculation can carry in the reconstruction problem.

A Hawking calculation can establish a radiation channel. A stress-energy calculation can contribute to the source or flux ledger. A two-point function can contribute to correlation data. A detector model can produce operational records. These are all legitimate exterior or semiclassical witnesses. But whether they preserve a threatened standing class depends on the relation between the outgoing data and the old reference class.

This is why the theorem is compatible with both information-loss and information-preserving semiclassical discussions. If the semiclassical exterior state lacks the required distinctions, then the old exterior scope loses standing. If a corrected or extended calculation supplies correlations, the certificate status changes. The AASC theorem does not choose the dynamics; it fixes the status of the reconstruction claim.

12.18 63. Relation to quantum gravity

A complete quantum-gravity theory may change the black-hole story at several levels. It may modify the near-horizon state, replace the classical interior, resolve singularities, provide boundary reconstruction, introduce new topological sectors, or define a nonperturbative Hilbert space. AASC does not exclude any of these routes.

The only requirement is role declaration. If the theory changes the metric projection, it changes anchors or tensor load. If it supplies a boundary algebra, it may supply a certificate. If it replaces the interior with microstate geometries, it must state the exterior standing of the distinctions. If it resolves singularities, it must state whether the resolution is a high-curvature tensor load, endpoint certificate, or larger-scope continuation.

Thus quantum gravity can satisfy the theorem rather than conflict with it. The theorem is a sieve: it distinguishes completed certificate structures from promises that a future theory will restore standing. A promise is not a certificate. A declared map is.

12.19 64. Relation to the standard black-hole literature

The standard black-hole literature separates naturally into several layers that match the role distinctions used in this paper. The classical geometry layer supplies exact and idealized metric representatives: Schwarzschild for the vacuum spherical exterior, Reissner-Nordstrom for charged electrovac response, Kerr for stationary rotating response, and Kerr-Newman for combined mass, charge, and angular momentum. AASC uses these as metric representatives inside a licensed projection. It does not ask them to carry exterior reconstruction certificates unless the surrounding model supplies such a role.

The singularity and horizon literature supplies the diagnostic and causal layer. Penrose-type singularity theorems show that under suitable hypotheses gravitational collapse leads to geodesic incompleteness. Hawking-Ellis style global methods clarify causal structure, horizons, and incompleteness. AASC preserves these as metric and causal results. It adds only the claim-status distinction: geodesic incompleteness and curvature pathology are diagnostic of metric exhaustion; they are not primitive terminal ontology.

The black-hole mechanics and thermodynamics literature supplies the ledger layer. The Bardeen-Carter-Hawking laws, Bekenstein entropy, Hawking tem-

perature, and Page-style entropy behavior are central. AASC preserves their significance while separating thermodynamic, entropic, and reconstruction roles. The first law governs exteriorly licensed terms. The Page curve constrains entropy behavior. Neither is automatically a full standing-transport map unless the relevant reconstruction relation is supplied.

The information-paradox literature supplies the reconstruction layer. Complementarity, holography, AdS/CFT, firewalls, islands, fuzzballs, final-state projection, remnants, and baby universes are not treated as mere opinions. They are programs that attempt to supply a certificate, change the scope, add horizon tensor load, replace the interior, or move retention into a larger domain. The AASC normal form classifies what each program must provide to count as same-scope exterior reconstruction.

The reference list therefore has two functions. The standard references locate the black-hole structures being classified. The AASC source manuscripts provide the role and admissibility machinery. The paper’s new contribution is not any single imported theorem. It is the black-hole reconstruction object, the Horizon Isolation Lemma, the induced partial-domain obstruction, and the application of the normal form across the standard black-hole problem space.

12.20 65. Role classification of common black-hole sentences

A polished black-hole theorem must be able to classify ordinary sentences without changing vocabulary. The following examples show the normal form in use.

“The black hole has mass M .” In an asymptotically flat exterior this is a mass/source-response claim, often tied to ADM or related asymptotic structure. In a radiating context it may be a Bondi-type ledger. It is not primitive interior substance.

“The black hole has charge q .” This is a gauge-boundary response claim fixed by electromagnetic field or flux witnesses. It does not reconstruct every charged object that fell in.

“The black hole has spin J .” This is an angular-momentum response claim. The physical witnesses include frame dragging, asymptotic rotational charge, stationary-axisymmetric structure, and the thermodynamic angular-velocity term. It is not microscopic spin ontology.

“The horizon hides the interior.” This is a metric/causal and witness-access claim. If used to claim permanent loss of standing, it triggers the horizon theorem. If a certificate is supplied, the case changes.

“The singularity destroys information.” This sentence is inadmissible unless a specific standing-bearing reference class, endpoint dynamics, and destruction mechanism are declared. A singularity by itself is a metric-exhaustion diagnostic.

“The information is in the radiation.” This is a certificate claim only if radiation observables or correlations are mapped to the threatened reference classes. Otherwise it is a slogan.

“The information is in the dual theory.” This is a holographic certificate claim only if the dual map preserves the exterior standing quotient. Otherwise it is larger-domain retention.

“The entropy counts microstates.” This is a microphysical state-counting claim. It requires a carrier, equivalence relation, counting measure, and relevance to exterior standing if used in the reconstruction problem.

“The remnant stores everything.” This is a storage claim. It becomes a reconstruction claim only with an exterior remnant map.

“The island solves the paradox.” This is a reconstruction-domain claim. It succeeds when the island prescription factors through the declared exterior algebra and preserves the target quotient.

These sentence classifications reveal the manuscript’s core discipline. The paper does not forbid ordinary black-hole language. It assigns each sentence to the role that makes it physically meaningful and blocks the hidden promotion of one role into another.

12.21 66. Why the normal form is not merely taxonomy

A taxonomy lists categories. The black-hole normal form does more: it constrains permissible transitions between categories. A metric descriptor cannot become an exterior witness merely by being named. An entropy ledger cannot become an interior microstate ontology without a counting and witness map. A global state cannot become exterior reconstruction without Rec_G . A hidden remnant label cannot become standing preservation without exterior access. These are not labels placed after the fact; they are transition rules.

The main transition rule is the no-totalization theorem. Once the Horizon Isolation Lemma removes old-scope exterior images for threatened classes, the transport operator is partial. The theory cannot move from partiality to totality without naming the source of the new definedness. That source determines the role: certificate, leakage, quotient, scope change, selector, repair, carrier laundering, or bookkeeping.

This is why the normal form has predictive force at the structural level. It predicts that any successful same-scope solution must expose its certificate. It predicts that a hidden storage route without exterior map fails as exterior reconstruction. It predicts that classical singularities cannot serve as terminal information-processing objects without richer dynamics. It predicts that a purely thermodynamic relation cannot by itself settle reference recovery. These are not numerical predictions; they are admissibility predictions.

The normal form also improves comparison among programs. It avoids asking whether holography, fuzzballs, or remnants are “true” in the abstract. It asks which standing role each program claims to fill and whether it fills it. This makes partial successes legible. A fuzzball count may explain entropy. A holographic code may reconstruct a subalgebra. A remnant model may preserve a coarse parameter. The normal form reports each success at its proper level.

12.22 67. Limits of scope and legitimate extensions

The present theorem is deliberately fixed-domain. A legitimate extension can alter the domain. Quantum gravity can change the metric projection. Holography can replace the naive exterior algebra with a boundary algebra. Islands can enlarge the reconstruction domain. Fuzzball models can replace the classical interior. Remnant models can declare a late object. Modified gravity can add higher-curvature tensor load. None of this is rejected.

What the extension cannot do is claim to preserve the old theorem object while silently moving its anchors. If the exterior algebra changes, say so. If a new tensor sector is introduced, say so. If a final-state boundary condition is added, say so. If only coarse recovery is claimed, say so. This is the no-backflow posture of the manuscript.

The result is compatible with progress. A future theory may supply a complete certificate for evaporating black holes. If so, it will not refute the no-totalization theorem. It will satisfy the certificate alternative. A future theory may show that the classical horizon is replaced by fuzzball microstructure. If so, the old permanent horizon object is gone. A future theory may show that islands are the correct reconstruction domains. If so, the exterior algebra has been enriched by a certified prescription.

Thus the manuscript is not conservative in the sense of preserving old ignorance. It is conservative in claim-status discipline. It keeps each theorem attached to the scope in which its terms stand.

13 Part XIII - Black-Hole Normal Form

13.1 68. Normal-form classifier

The black-hole normal form is now earned. In the main text the roles are grouped into five families.

1. **Licensed exterior and metric roles:** exterior observables, licensed metric descriptors, mass response, charge response, and angular-momentum response.
2. **Boundary and reconstruction roles:** horizon constraints, radiation or leakage channels, standing-transport certificates, and reconstruction or correlation claims.

3. **Diagnostic and ledger roles:** singularity or failure diagnostics and thermodynamic ledgers.
4. **Extension roles:** admissible richer-scope extensions and explicit scope changes.
5. **Failure or inert roles:** hidden selectors, post-hoc repairs, carrier laundering, hidden interior fact upgrades, skin, and fail-closed claims.

The full nineteen-role classification table is collected in Appendix E. The grouping keeps the main theorem readable while preserving the fine-grained classification categories used for counterexamples.

Theorem 15 - Black-hole normal form. Every black-hole claim in a fixed exterior black-hole reconstruction scope, evaluated under the AASC role grammar used in this manuscript, has one of the listed role images, up to bookkeeping overlap. No twentieth same-scope role remains.

Proof. A claim either affects exterior witnesses, metric descriptors, horizon boundary typing, singularity diagnostics, parameter ledgers, thermodynamic ledgers, radiation/leakage channels, certificate maps, reconstruction/correlation status, or scope. If it affects none of these, it is skin or bookkeeping. If it selects an outcome without witness support, it is a selector. If it makes an old undefined transport defined after loss, it is repair. If it treats a derived item as primitive authorization, it is carrier laundering. If it imports hidden interior standing, it is an interior fact upgrade. If it cannot be licensed, it fails closed. Anchor/tensor/skin discipline and the partial-domain obstruction leave no fourth governance locus and no twentieth black-hole role.

13.2 69. Counterexample normal forms

A counterexample to the horizon theorem must exhibit

$$S_{\text{BH}}, \quad Q_{\text{ext}}, \quad A_{\text{wit}}, \quad \textit{Cont}, \quad T_H^*$$

with a permanent horizon, no leakage, no certificate, no non-sink quotient, preserved old scope, totalized T_H , no selector, no repair, no scope change, no carrier laundering, no bookkeeping, and no exhaustion. The no-totalization theorem says this object cannot exist in the declared scope.

A counterexample to the normal form must exhibit a twentieth black-hole role that is not an exterior observable, metric descriptor, horizon constraint, singularity diagnostic, parameter response, thermodynamic ledger, radiation channel, certificate, reconstruction claim, richer-scope extension, scope change, selector, repair, carrier laundering, hidden interior upgrade, skin, or fail-closed claim. Any such alleged role either moves an anchor, adds tensor content, changes scope, or changes nothing.

Arc handoff summary

This orientation paper leaves the reader with a fixed black-hole comparison object and no endpoint conclusion. Its final export is the disciplined exterior reconstruction target: declared witnesses, standing quotient, continuation relation, parameter and thermodynamic ledgers, boundary typing, certificate status, and a no-selector/no-repair rule. The next paper may use that package to prove an exterior obstruction; it may not treat the hidden carrier, AMetric endpoint, or fresh-domain capstone as already available.

14 Part XIV - Conclusion

Series-position conclusion. This orientation paper closes by returning only the role-typed black-hole package to the arc. It does not export endpoint status or fresh-domain status. Its polished series function is to make later handoffs legible: the exterior theorem tests the reconstruction burden fixed here; the interior theorem uses the standing break exported by that exterior test; the homology and capstone papers enter only after those intervening results.

14.1 70. Result

Given the AASC admissibility and standing constraints, the paper began with a distinction between a classical event horizon and an admissible horizon reconstruction object. It then fixed an exterior black-hole reconstruction scope, constructed the exterior standing quotient and witness algebra, introduced threatened exterior reference classes, proved the Horizon Isolation Lemma, induced the partial horizon standing-transport operator, and proved that same-scope totalization is impossible without certificate, leakage, non-sink quotient, or scope change.

This proof does not invalidate classical black-hole metrics. It preserves them as metric representatives inside a licensed projection. What it denies is the further claim that a permanent uncertified horizon can preserve exterior standing as a continuing same-scope reconstruction object.

Within that conditional AASC theorem scope, the downstream classifications follow. Singularities are failure signatures of metric representation, not terminal objects. Mass, charge, and angular momentum are exterior response roles, not hidden interior facts. The first law is preserved as a thermodynamic ledger. Entropy is not an ontology generator. Information is not primitive. Global retention becomes exterior recovery only through a certificate. Holography, islands, complementarity, remnants, fuzzballs, final-state projection, and baby universes are evaluated by the certificate or scope-change structures they supply.

The admissible black hole is neither a primitive breakdown of physics nor a hidden interior ontology. It is a licensed metric-interior limit regime whose exterior standing, horizon boundary, parameter response, thermodynamic ledger,

and reconstruction certificates must be kept distinct. Once those distinctions are enforced, uncertified permanent horizons fail as continuing same-scope reconstruction objects; singularities lose ontic endpoint status; entropy and information remain indispensable ledgers without becoming primitive; and quantum-gravity programs are evaluated by the exterior-standing certificates they supply.

15 Appendix A - Formal dictionary

Symbol	Meaning
S_{BH}	Fixed exterior black-hole reconstruction scope
E_{ext}	Exterior admissible description domain
BH	Candidate horizon or horizon-like boundary
R_{ext}	Exterior standing-bearing reference classes
W_{ext}	Exterior witness family
A_{wit}	Exterior witness algebra
Q_{ext}	Exterior standing quotient
$Cont$	Admissible continuation relation
Env	Envelope ledger of physical assumptions
$Param$	Mass, charge, and angular-momentum response ledger
$Therm$	Thermodynamic ledger
$Cert$	Certificate status field
X_H	Source-horizon-exterior carrier
$E_{\partial H}$	Horizon boundary trace envelope
E_{int}	Horizon-isolated carrier relative to exterior scope
T_H	Partial horizon standing-transport operator
T_H^*	Attempted totalization of T_H
$Cert_H$	Standing-transport certificate family
Rec_G	Exterior recoverability certificate induced by a larger-domain state
M_{ext}	Exterior mass/source-response role
q_{BH}	Black-hole charge role
J_{ext}	Exterior angular-momentum role

16 Appendix B - Claim status and theorem cluster inventory

16.1 Claim-status ledger

Claim	Status
Classical GR black-hole solutions are useful metric representatives	Preserved
Semiclassical Hawking radiation is denied	Not claimed
AASC derives Hawking temperature, greybody factors, or evaporation time	Not claimed
AASC derives a Page curve	Not claimed
AASC solves quantum gravity dynamically	Not claimed
AASC denies inaccessible interiors in all larger domains	Not claimed
AASC classifies same-scope exterior black-hole reconstruction	Main claim
A permanent uncertified horizon is a standing sink	Main theorem, after horizon isolation is proved
Singularity is an ontic endpoint	Denied
Singularity is an exhaustion or failure signature of metric representation	Main theorem
Mass, charge, and angular momentum are hidden interior facts	Denied
Mass, charge, and angular momentum are exteriorly licensed response roles	Main theorem
Black-hole entropy proves truth-apt interior microstate inventory	Denied
Entropy is an admissible thermodynamic or reconstruction ledger	Preserved
Information is primitive ontology	Denied
Information is bookkeeping over standing-bearing carriers and encodings	Preserved
Holography may be admissible	Yes, if it supplies exterior anchor transport
Complementarity may be admissible	Yes, if it supplies redescription-stable correspondence
Remnants may be admissible	Yes, if they expose an exterior remnant map

Claim	Status
Baby universes are same-scope rescues	Denied unless an exterior certificate is induced
Final-state projection is admissible by default	Denied; selector risk unless declared as physical boundary data
Fuzzballs or microstate geometries are rejected	Not claimed; classified by certificate behavior
The paper gives the black-hole normal form under AASC	Main endpoint

The theorem inventory is organized into six proof clusters. The organization is intentionally sparse: each cluster names a proof stage rather than listing every corollary.

16.1.1 Cluster I - Regime and metric-projection foundation

1. **Black-hole target-regime instantiation.** Black-hole descriptions instantiate the AASC target class because they rely on exterior observables, retained identity, metric comparison, causal and temporal continuation, mass/charge/angular-momentum ledgers, thermodynamic accounting, radiation histories, and reconstruction claims.
2. **Metric-projection floor.** Classical black-hole geometry is evaluated inside a faithful metric projection of the AASC gravity role. A metric representative may be valid without becoming a primitive admissibility boundary.
3. **Black-hole reconstruction object.** A black-hole regime is a fixed-scope object containing exterior scope, metric projection, horizon boundary typing, exterior witness algebra, standing-bearing exterior reference classes, parameter ledgers, thermodynamic ledgers, continuation relation, radiation or leakage status, and certificate status.

16.1.2 Cluster II - Horizon obstruction

4. **Exterior standing quotient.** Exterior black-hole standing is the quotient of exterior reference classes preserving witness, continuation, boundary-access, and admissibility-status profiles.
5. **Exterior witness algebra.** Same-scope standing recovery must factor through the declared witness algebra, leakage channel, certificate map, continuation relation, or non-sink quotient.
6. **Horizon Isolation Lemma.** A permanent horizon with no leakage, boundary trace, radiation witness, anchor map, remnant map, or non-sink quotient leaves threatened exterior standing classes without an admissible image in the old exterior witness algebra.
7. **Partial-domain induction.** Once the Horizon Isolation Lemma holds, the standing-transport operator is partial on the old carrier: it is not

defined on the horizon-isolated interior envelope.

8. **No-totalization theorem.** No same-scope operator $T_H^* : X_H \rightarrow R_{\text{ext}}$ can extend T_H across the old uncertified interior while preserving the old exterior quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status.
9. **No uncertified permanent-horizon reconstruction.** An uncertified permanent horizon is not an admissible continuing same-scope standing-preserving exterior reconstruction object. The admissible alternatives are leakage, evaporation, certified exterior reconstruction, holographic anchor transport, complementarity with correspondence, a non-sink quotient, richer scope, or explicit exterior-successor exhaustion.

16.1.3 Cluster III - Singularity and high-curvature diagnostics

10. **Singularity non-object theorem.** A black-hole singularity is an exhaustion signature of fixed metric articulation, not a physical object, terminal fact, or primitive admissibility endpoint.
11. **High-curvature route classification.** High-curvature endpoint claims require declared higher-response tensor load, effective scope, lawful leakage or radiation channel, endpoint certificate, non-sink quotient, or richer-scope theory. Hidden endpoint repair is inadmissible.

16.1.4 Cluster IV - Exterior parameter roles

12. **Mass role theorem.** Black-hole mass is an exterior gravitational/source/deformation-response ledger, not primitive interior substance.
13. **Charge role theorem.** Black-hole charge is a gauge-compatible exterior electromagnetic response fixed by boundary or flux witnesses.
14. **Angular-momentum role theorem.** Black-hole spin, technically angular momentum, is an exterior rotational, asymptotic, and frame-dragging response role, not a hidden interior rotation ontology.
15. **Kerr-Newman normal form.** The stationary charged rotating exterior is admissible as a role package $(M_{\text{ext}}, q_{\text{BH}}, J_{\text{ext}})$ plus metric, horizon, thermodynamic, radiation, and certificate data.

16.1.5 Cluster V - Thermodynamics and information

16. **Thermodynamic ledger theorem.** Black-hole temperature, entropy, area, surface gravity, and first-law terms are admissible exterior thermodynamic ledgers.
17. **Entropy non-promotion theorem.** Entropy does not by itself license truth-apt interior microstate ontology or standing transport.
18. **Information non-primitivity theorem.** Information-loss claims must specify carriers, encodings, exterior witnesses, and reconstruction maps.
19. **Global-retention separation theorem.** A global state, dual representation, Page-curve result, remnant Hilbert space, island construction, or

quantum-gravity completion preserves exterior standing only if it induces an exterior recoverability certificate.

16.1.6 Cluster VI - Program classification and normal form

20. **Program classification theorem.** Semiclassical evaporation, holography, islands, complementarity, firewalls, remnants, fuzzballs, final-state projection, and baby-universe proposals are classified by whether they supply certificate structure, change scope, add declared tensor load, or fail same-scope standing preservation.
21. **Black-hole normal-form theorem.** Every black-hole claim in the fixed scope has one of the listed role images, and no twentieth same-scope role remains.
22. **Counterexample normal form.** A counterexample must preserve the old exterior quotient, witness algebra, continuation relation, parameter ledger, thermodynamic ledger, and certificate status while totalizing horizon transport without certificate, leakage, non-sink quotient, selector, repair, carrier laundering, scope change, exhaustion, or bookkeeping.

17 Appendix C - Source-use ledger

Source family	Imported into this manuscript	Not imported
Horizon obstruction	exterior scope, witness algebra, certificate burden, partial transport, no-totalization	Hawking spectrum or Page curve
Einstein projection	metric floor and high-curvature route classification	all gravitational dynamics or numerical constants
Realized physical interior	physical role stack	full Standard Model
Mass/inertia	mass as deformation/source response	parameter closure black-hole mass values
Information non-primitivity	no informational ontology	denial of information theory
Measurement closure	record fixation and no primitive collapse/branching	collapse dynamics
Arrow of time	standing monotonicity, entropy/record/cosmological arrow discipline	thermodynamic mechanism
Failure signatures	singularity diagnostic	singularity resolution
Scale-standing	no Planck-label or scalar representative promotion	hierarchy numerical value

18 Appendix D - Proof dependency graph

$K_{\text{AASC}} \rightarrow \text{RPI} \rightarrow \text{AASC Gravity} \rightarrow \text{Metric Projection} \rightarrow S_{\text{BH}}$
 $\rightarrow (Q_{\text{ext}}, A_{\text{wit}}) \rightarrow \text{Horizon Isolation} \rightarrow \text{Partial } T_H$
 $\rightarrow \text{No Totalization} \rightarrow \text{No Uncertified}$
 $\rightarrow \text{Permanent Horizon} \rightarrow \text{Normal Form.}$

19 Appendix E - Standard black-hole role dictionary

19.1 Full normal-form classification table

Family	Role image
Licensed exterior/metric	exterior observable
Licensed exterior/metric	licensed metric descriptor
Licensed exterior/metric	mass response datum
Licensed exterior/metric	charge response datum
Licensed exterior/metric	angular-momentum response datum
Boundary/reconstruction	horizon constraint
Boundary/reconstruction	radiation or leakage channel
Boundary/reconstruction	standing-transport certificate
Boundary/reconstruction	reconstruction or correlation claim
Diagnostic/ledger	singularity or failure diagnostic
Diagnostic/ledger	thermodynamic ledger
Extension	admissible richer-scope extension
Extension	scope change
Failure/inert	hidden selector
Failure/inert	post-hoc repair
Failure/inert	carrier laundering
Failure/inert	hidden interior fact upgrade
Failure/inert	skin
Failure/inert	fail-closed claim

Standard object	AASC role
Schwarzschild metric	vacuum exterior mass-response representative
Reissner-Nordstrom metric	electrovac mass/charge representative
Kerr metric	stationary rotating exterior representative
Kerr-Newman metric	mass/charge/angular-momentum role package

Standard object	AASC role
Event horizon	metric/causal representative; not automatically reconstruction certificate
Apparent/trapping horizon	horizon boundary role requiring declared scope
Singularity	exhaustion/failure diagnostic
Surface gravity	temperature ledger
Area	entropy/thermodynamic ledger
Hawking radiation	leakage/radiation channel candidate
Page curve	entropy/correlation evidence; certificate only with reconstruction map
Holographic dual	certificate candidate if anchor transport is supplied

20 Appendix F - Program comparison matrix

Program	Contribution	Success condition	Failure mode	Certificate form
Semiclassical evaporation	radiation/leakage	radiation carries standing witnesses	thermal ledger without reconstruction	radiation-reference map
Holography	boundary dual	preserves Q_{ext} through boundary algebra	encoding assertion without map	boundary reconstruction map
Islands	reconstruction domain	factors through exterior algebra	scope change as recovery	algebraic reconstruction map
Complementarity	observer consistency	stable correspondence	observer-scope drift	exterior-infalling map
Firewall	horizon tensor load	declared physical horizon load	illegal bookkeeping demand	horizon tensor model
Remnant	late storage	exterior remnant map	hidden compression sink	remnant-witness map
Fuzzball	microstate replacement	exterior microstate relation	hidden labels	exterior microstate witnesses
Final-state projection	endpoint boundary	declared physical anchor	post-hoc selector	licensed final-state condition
Baby universe	larger-domain retention	exterior image induced	scope transport	branch-to-exterior map

21 Appendix G - Hostile-objection ledger

Objection	Response
Classical GR has event horizons	Preserved. The theorem targets admissible horizon reconstruction, not metric existence.
This is definition-driven	The Horizon Isolation Lemma proves the physical bridge before operator partiality is introduced.
Holography solves this	It may, if it supplies the required exterior certificate.
The interior exists in GR	Metric extension is preserved; same-scope exterior-standing facthood is separate.
Entropy counts microstates	Only in a declared counting ledger with carrier, equivalence relation, and witness role.
Unitarity preserves information	Global retention requires an exterior recoverability certificate.
Remnants store everything	Capacity is not standing; an exterior remnant map is required.
Spin is underdeveloped	The paper treats black-hole spin as angular momentum exterior response, not microscopic spin ontology.
This lacks dynamics	Correct. It is a fixed-domain reconstruction theorem; dynamics may realize the admissible routes.

22 Appendix H - Formalization boundary

This manuscript does not report a paper-specific Lean formalization of the black-hole reconstruction theorem itself. Its kernel-level infrastructure, however, is downstream of the Lean 4 formalization of the non-degenerate construction kernel and admissibility interface. That formalization treats *Adm*, *Stand*, *Ref*, and *Irr* as a fixed-domain closure structure and records fixed-domain kernel uniqueness, non-derivability from below, no faithful lower generator, and no deeper same-domain invariant. The natural next mechanization boundary for this paper is the fixed exterior reconstruction object, the exterior standing quotient, the exterior witness algebra, the Horizon Isolation Lemma as a predicate-level assumption package, the partial horizon standing-transport operator, the no-totalization theorem, route exhaustion, and the normal-form classifier.

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